Poultry Diseases
Overview, Anatomy

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Outlook

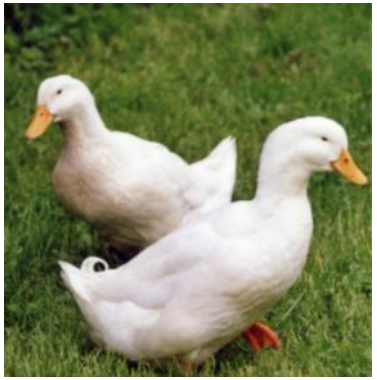
- 1. Poultry
- 2. Anatomy
- 3. Histology

Poultry

- Poultry are domesticated birds kept by humans for their **eggs**, their **meat** or their **feathers**.



Meat
Egg
Feather



Meat
Egg
Feather
Down(絨)
Foie gras



Meat
Egg
Feather
Down
Foie gras

Wikipedia



Turkey



Ostrich



Pigeon



Quail

- Turkey Europa, America
- Ostrich *Red meat
- Pigeon France, USA, North Africa, Asia
  (Not “Gaming pigeon”)
- Quail Japan, South East Asia

Wikipedia

- ZW sex-determination system

  - Chicken / Birds

  - **Male = ZZ**
  - **Female = ZW**

  - **Females** determine the sex of the offspring

  - Length: Z > W

  - Chromosome number:
    - Chicken: **78** (2n)
    - Turkey/ Duck/ Bird: **80** (2n)

Mammals

Male = XY
Female = XX

Males determine the sex of the offspring

Length: X > Y

Poultry vs. Mammal

- Feathers instead of fur
- No teeth
- Lay eggs
- Float and fly
- Excrete waste through one orifice only
- do not have sweat glands

Poultry vs. Mammal

- Unlike mammals, birds **lack a diaphragm** to inflate and deflate the lungs
- Birds have **air sacs** located in their neck and body cavity that inflate their lungs
  - Gas exchange occurs in the lungs and the air sacs function to move air in and out of the respiratory system

- Chicken - Breeding

Jungle Fowl (原雞)





Meat production



Egg production



Purebred

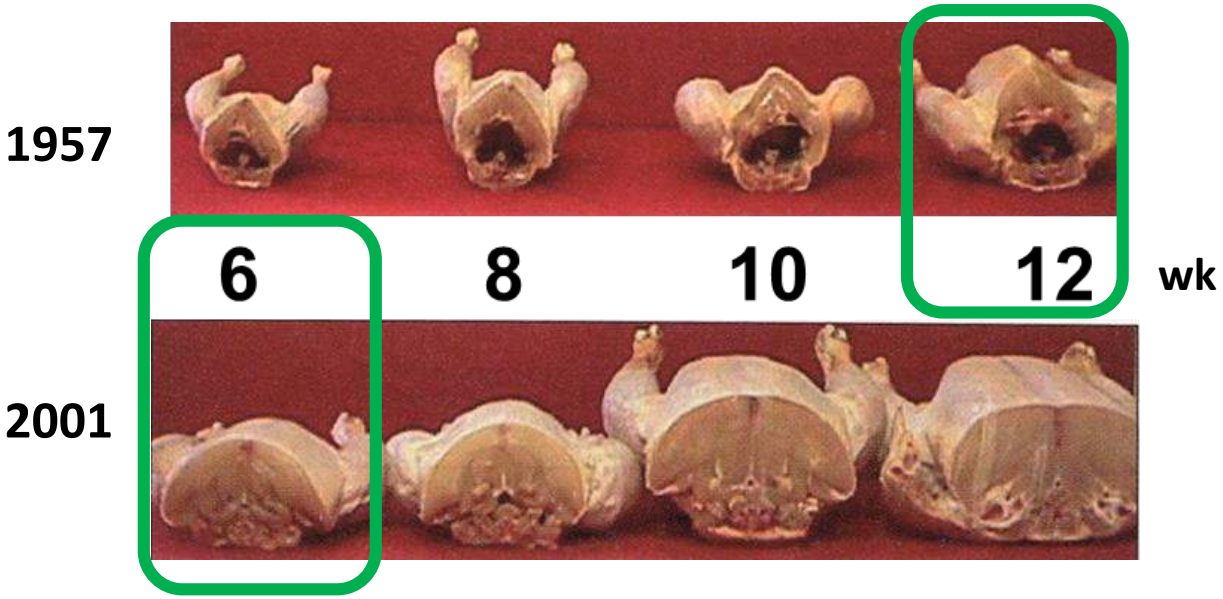
- Different breeds use different breeding strategies and different immunities plans.
- 雞在分類學上是動物界 (Animalia)–脊索動物門 (Chordata)–鳥綱 (Aves)–鶉雞目(Galliformes)–雉科 (Phasianidae) –雞屬 (Gallus)

Wikipedia

- Breeding goals – Broiler（肉雞、白肉雞）

Ross
Arbor Acres
Cobb
Hubbard
…etc





Broiler:

- Improved feed conversion
- Improved muscle development
- Less fat
- Bone stability
- Meat quality
- Improvement of general liveability
- better resistance against stressors (e.g. microbes)

飼料換肉率

2→1.4~1.5

Ross|Avigen



Arbor Acres
愛拔益加



ROSS
樂絲



Cobb
科寶

- Breeding goals – Laying hen（蛋雞）



Lohmann
Hy-Line
Hendrix
ISA



Chicken:
- Improved **feed conversion**
- Improvement of general liveability
- better resistance against stressors (e.g. microbes)

Egg:
- **Laying performance**
- Egg shell quality

Lohmann

LOHMANN TIERZUCHT

Hy-Line®

ISA

HENDRIX GENETICS



- Breeding goals – Colored Broiler（土雞、有色雞）





- Colorful Broiler:
- 紅羽、黑羽、皇金&其他
- **Appearance!!** (Body shape, Feather)
- Quality of meat production
- Hybrid vigor → general resistance against stressors (microbes)
- Genes and Inherited Diseases

Wikipedia



紅羽土雞



黃羽土雞



黑羽土雞



黑羽土雞

(屠宰家禽肉眼病理學圖譜)



閹雞



珍珠雞



鬥雞



烏骨雞

(屠宰家禽肉眼病理學圖譜)

台灣地區家禽產地每日交易行情表　2023

(單位：元 / 600公克)

日期	白肉雞	紅羽土雞北區	紅羽土雞中區	黑羽土雞公舍飼	黑羽土雞母舍飼	雞蛋	鵝肉(白羅曼)	番鴨(公)	土番鴨(75天)	鴨蛋(台南)
09/06	38.0	52.0	53.0	55.0	55.0	43.5	68.0	62.0	52.8	48.0
09/05	38.0	52.0	53.0	55.0	55.0	40.5	68.0	62.0	52.8	48.0

雞蛋價格以台北蛋商公會提供大運輸價格-3元估算

台灣地區家禽產地每日交易行情表　2024

(單位：元 / 600公克)

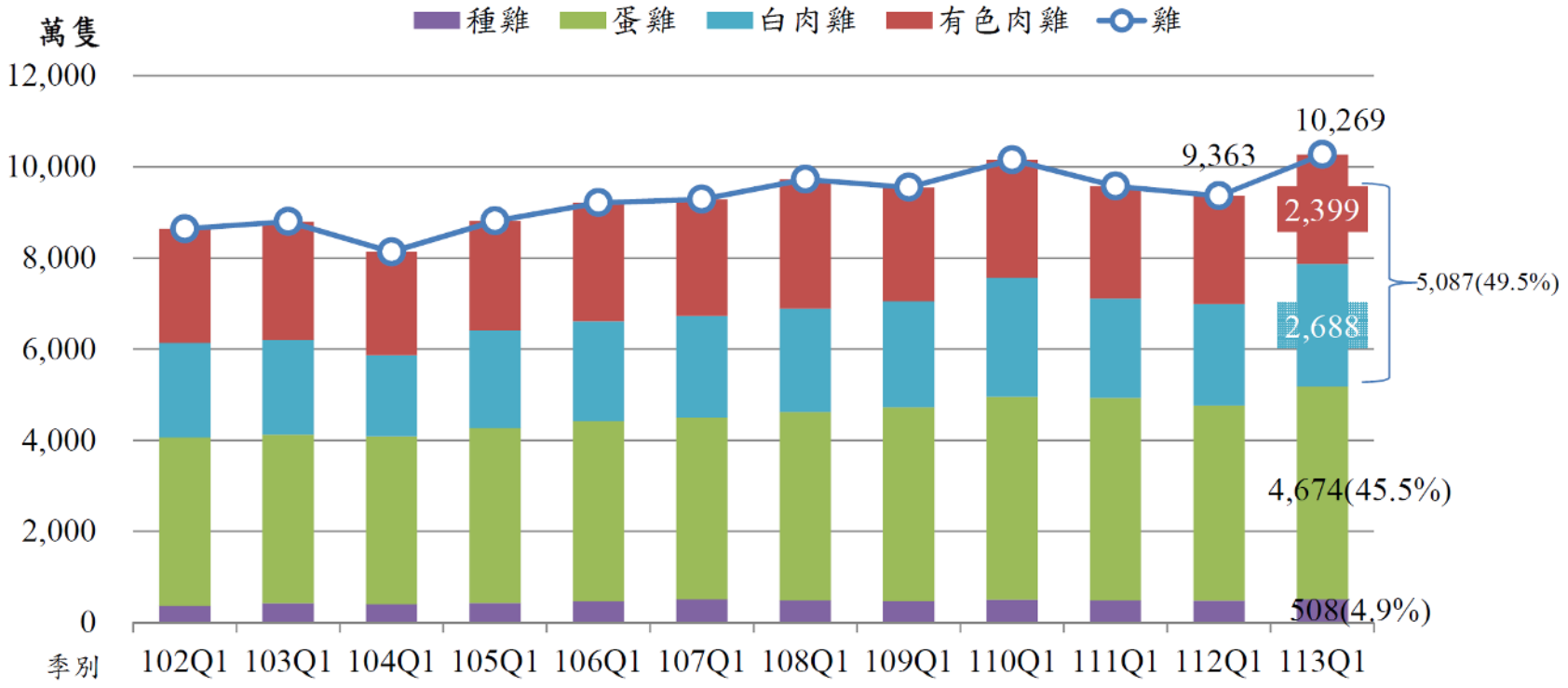
日期	白肉雞	紅羽土雞公北區	紅羽土雞母北區	紅羽土雞公中區	紅羽土雞母中區	紅羽土雞公南雞	紅羽土雞母南雞	黑羽土雞公舍飼	黑羽土雞母舍飼	雞蛋	鵝肉(白羅曼)	番鴨(公)	土番鴨(75天)	鴨蛋(台南)
09/09	34.0	53.0	53.0	54.0	54.0	54.0	54.0	52.0	52.0	28.5	69.0	62.0	52.8	42.0
09/08	34.0	53.0	53.0	54.0	54.0	54.0	54.0	52.0	52.0	28.5	69.0	62.0	52.8	42.0

雞蛋價格以台北蛋商公會提供大運輸價格-3元估算

113年第1季家禽在養情形(1/2)

2024年7月9日

養雞產業包括種雞、蛋雞及肉雞。113年第1季臺閩地區雞隻總在養量為10,269萬隻，較112年第1季增加8.8%。各類雞種在養情形：種雞在養508萬隻(占4.9%)、蛋雞在養4,674萬隻(占45.5%)、肉雞在養5,087萬隻(占49.5%)，其中白肉雞占肉雞總在養量52.8%、有色肉雞占47.2%。



資料來源：畜禽統計調查,農業部

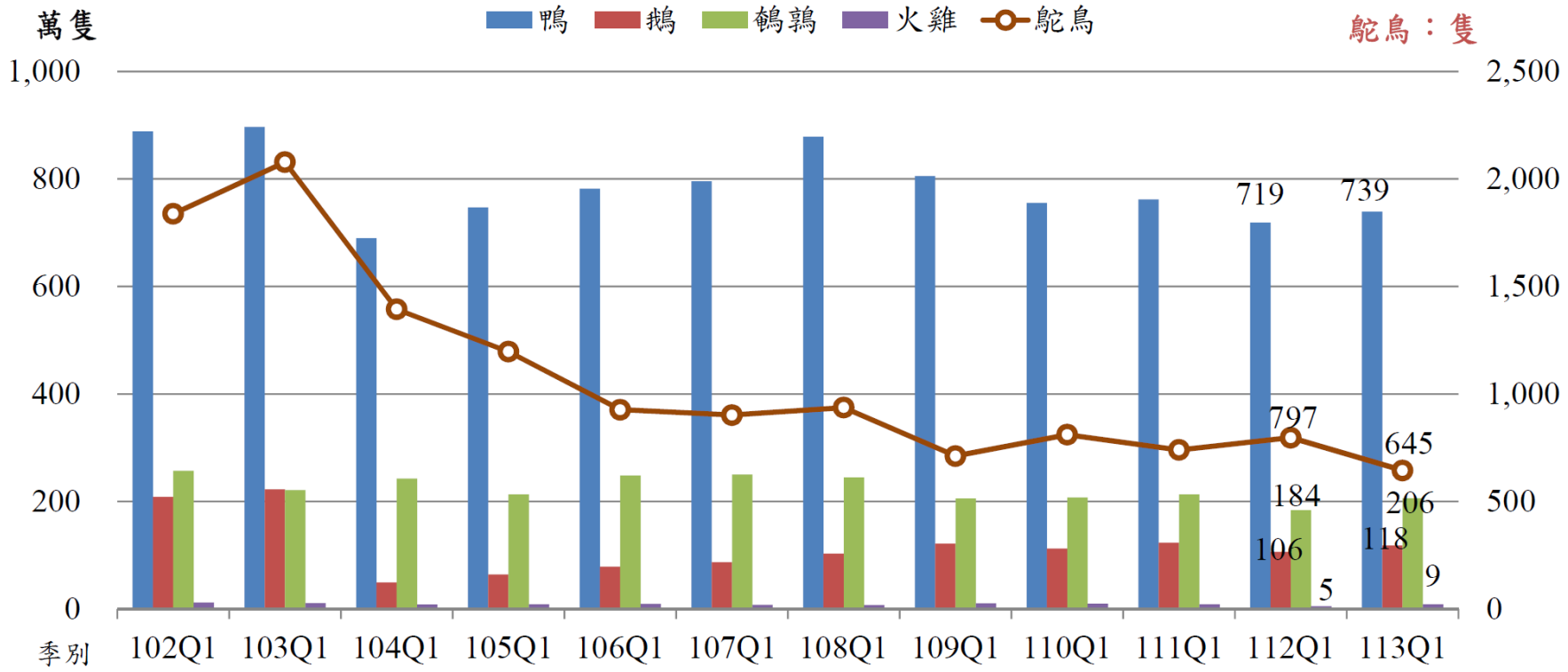
撰稿：李祐逸

審稿：傅敬懿代理組長/王旭昌代理執行長

113年第1季家禽在養情形(2/2)

2024年7月9日

113年第1季除雞以外之家禽在養量分別為鴨739萬隻、鵝118萬隻、鵪鶉206萬隻、火雞9萬隻、鴕鳥645隻。若與112年第1季比較，除鴕鳥在養量減少23.6%外，其餘均呈增加趨勢，鴨增加2.8%、鵝增加10.2%、鵪鶉增加10.8%、火雞增加41.7%。



資料來源：畜禽統計調查,農業部

中央畜產會

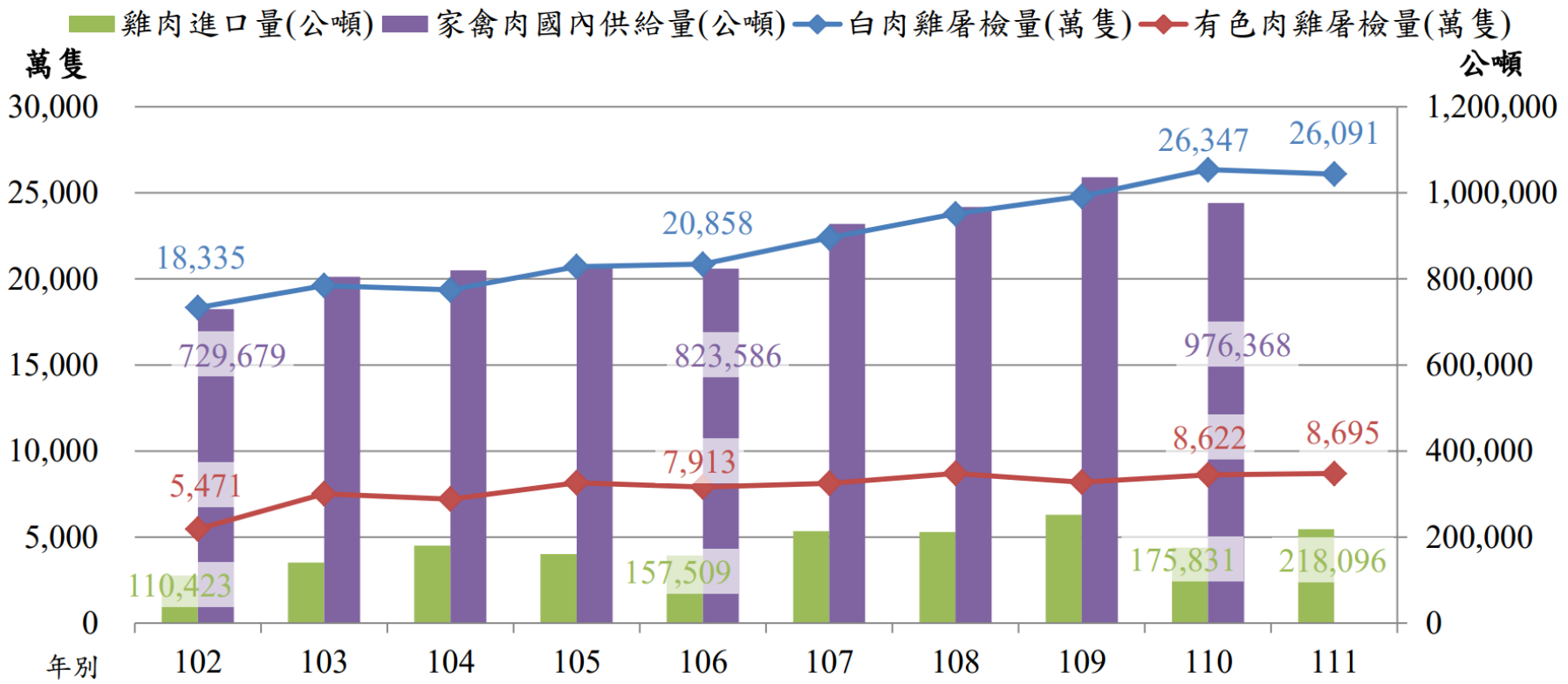
撰稿：李祐逸

審稿：傅敬懿代理組長/王旭昌代理執行長

近十年白肉雞與有色肉雞生產比較

近年國內家禽肉需求量逐年增加，國內白肉雞需求幅度較有色肉雞為高

111年國內白肉雞屠檢數量為26,091萬隻，較前110年減少0.97%、較106年增加25.09%；有色肉雞屠檢數量為8,695萬隻，較110年增加0.86%、較106年增加9.89%；111年雞肉進口量為218,096公噸、較110年增加24.04%、較106年增加38.47%；110年家禽肉國內供給量為976,368公噸、較106年增加18.55%。



資料來源：
「農業統計年報」、「農業貿易統計」，行政院農業委員會
「屠宰衛生統計」，動植物防疫檢疫局。

撰稿：李文棋
審稿：梁啟峰組長/王旭昌副執行長

中央畜產會

- Waterfowl

Goose



Duck



Black-faced spoonbill

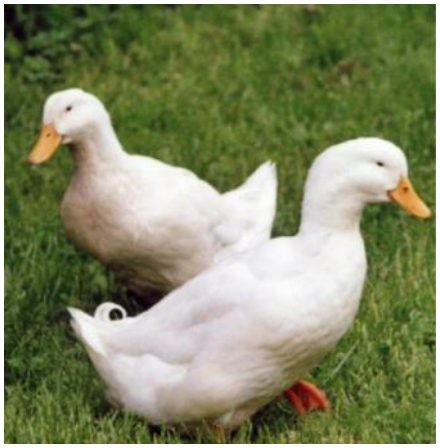


Common Teal (小水鴨)



Wikipedia

- Ducks



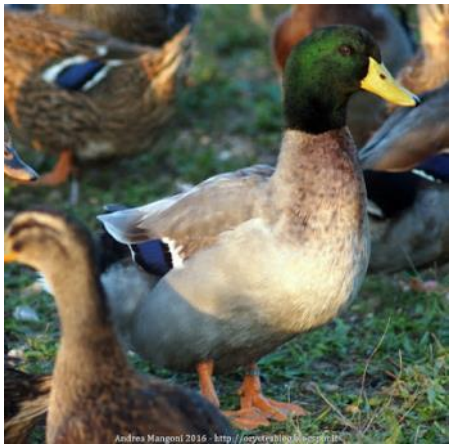
Pekin ducks
北京鴨



Tsaiya Duck
菜鴨



Muscovy duck
番鴨



Germanata Veneta
日耳曼綠頭鴨

- Products
  - Meat
  - Egg
  - Feather
  - **Down** (絨羽)
  - Foie gras

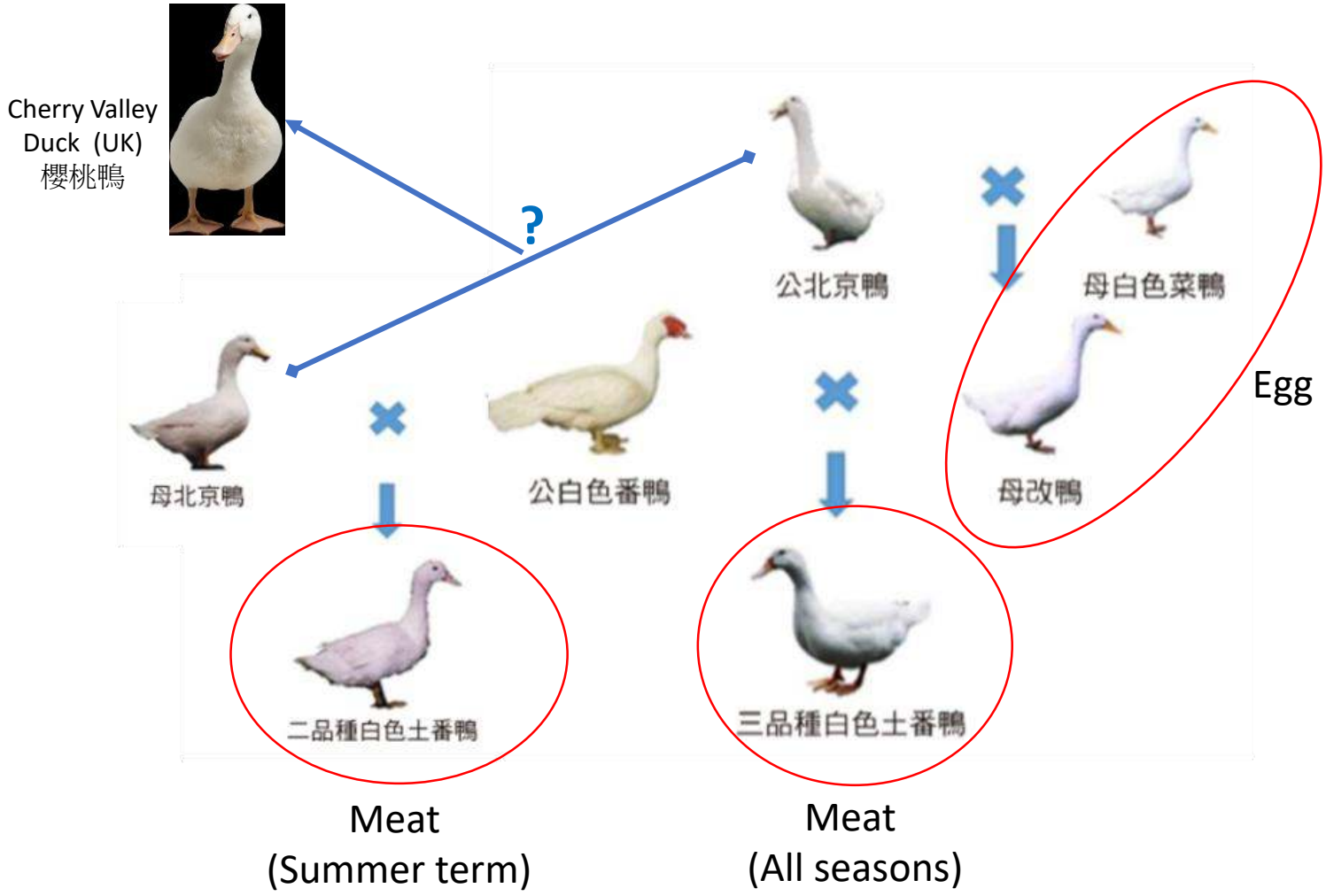




Down:
Waterproof
Warm

Wikipedia; Baden TS, 2014

- Breeding goals – Ducks



- Mule duck (土番鴨) = **NO** offspring.

Cherry Valley; Modified from Liu 2017

- Geese



Chinese goose 華鵝



White Roman goose
白羅曼鵝



Toulouse goose
法國 土魯斯鵝



Embdem goose
德國 艾姆登鵝

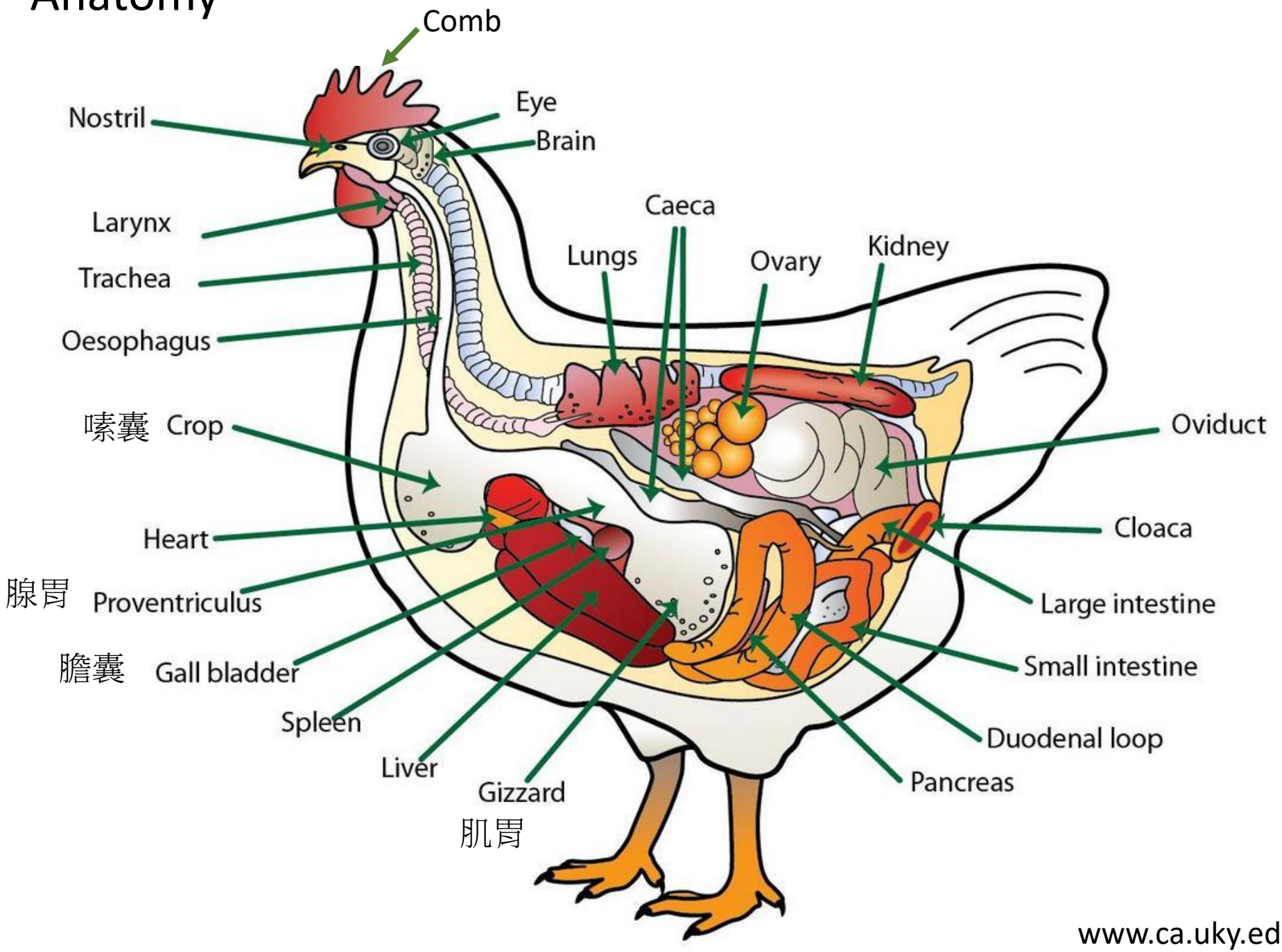
- Products
  - Meat
  - Egg
  - Feather
  - Down
  - **Foie gras** (鵝肝)



From Hungary

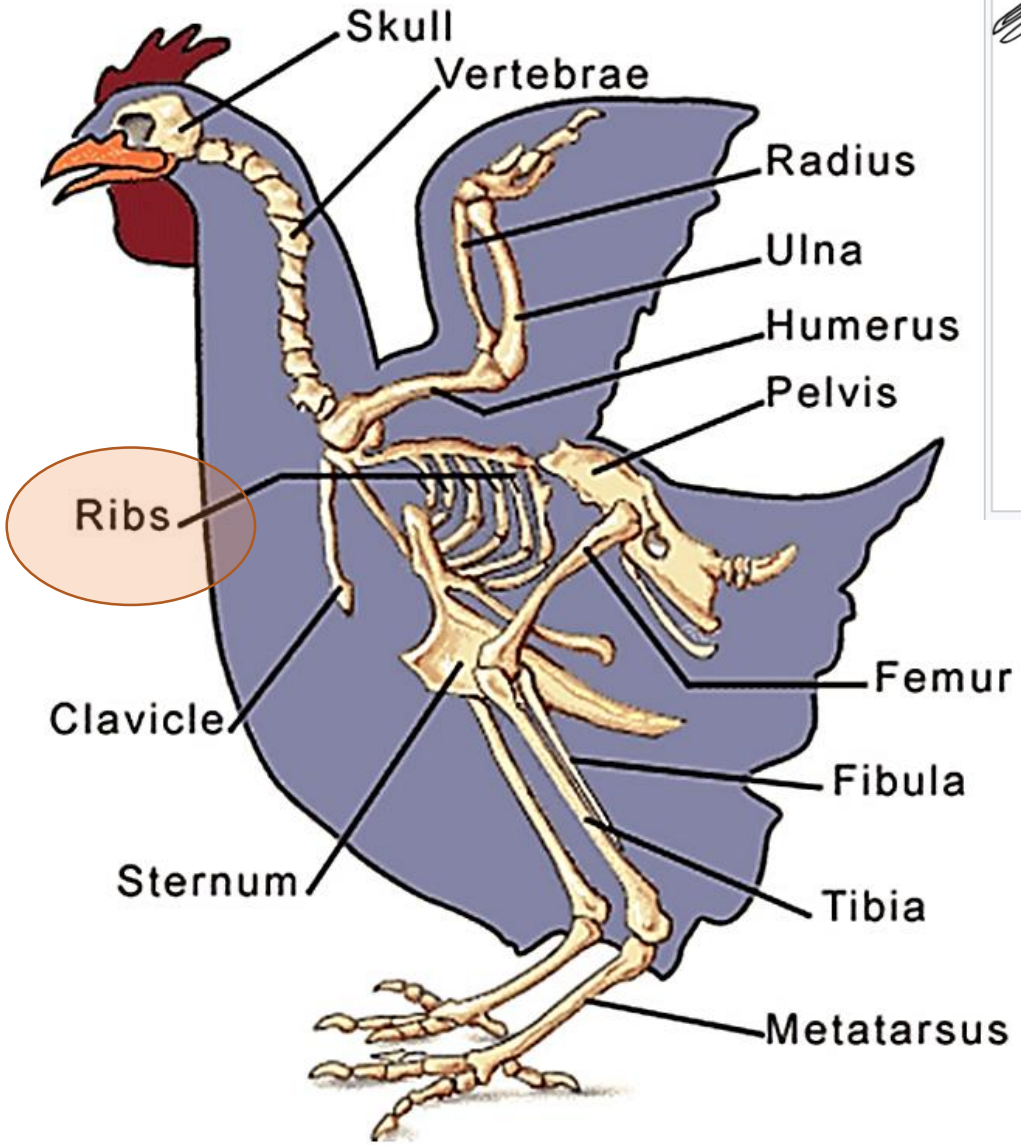
Wikipedia

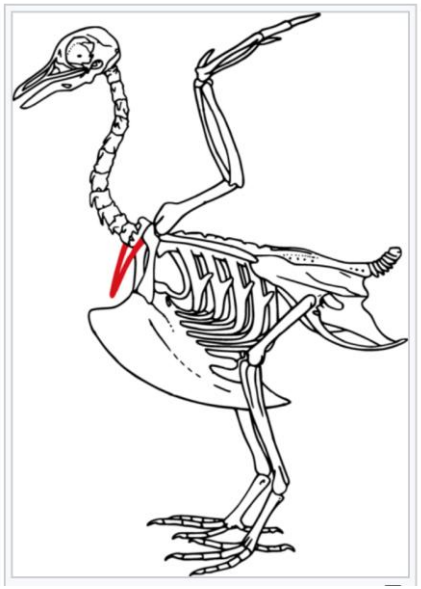
- Anatomy



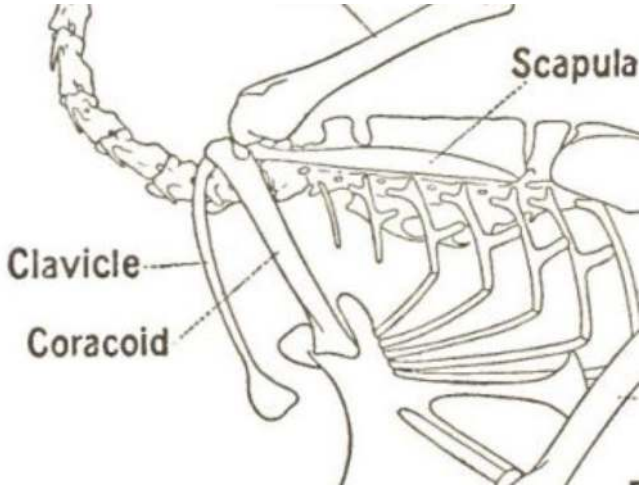
www.ca.uky.edu

- Skeletal system



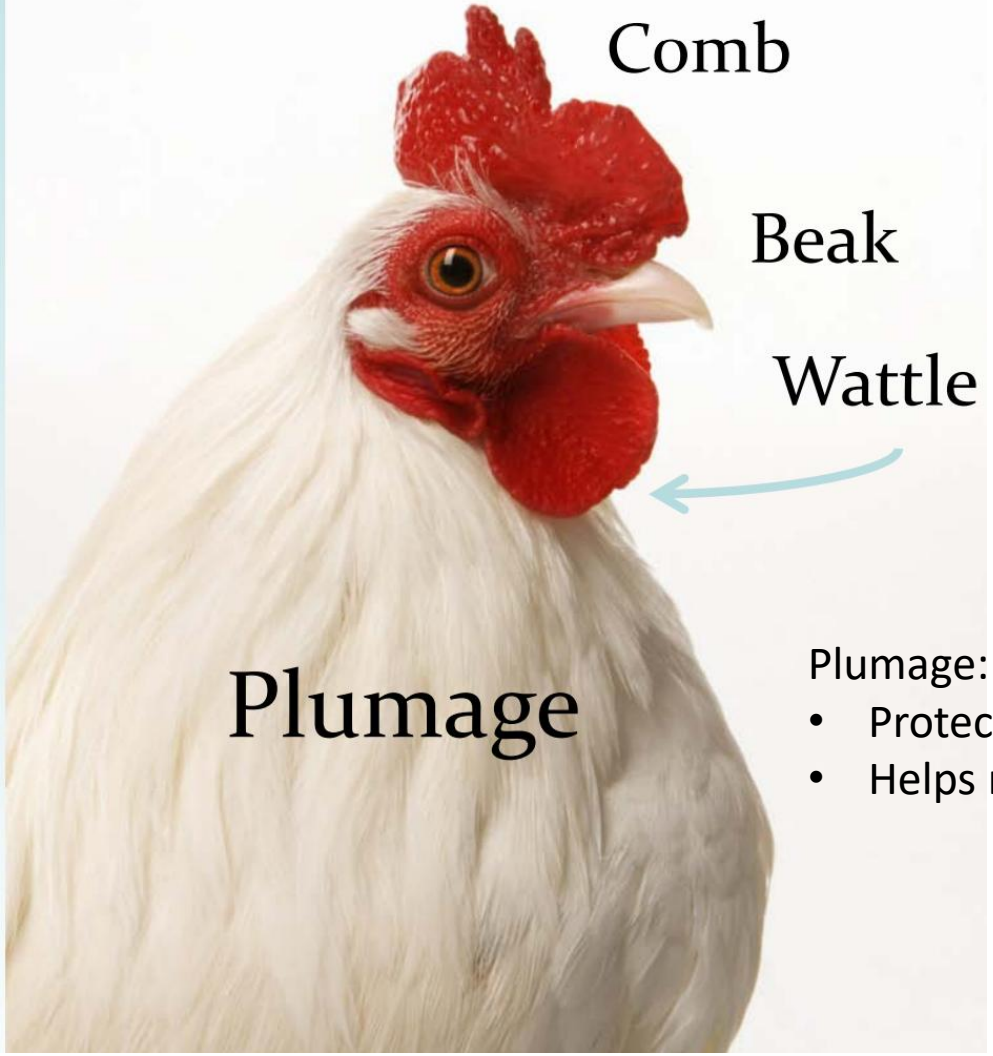


Hollow bone



www.ca.uky.edu

www.poultryhub.org



Plumage:

- Protects against cuts and bruises
- Helps regulate body temperature

Collect a tracheal swab

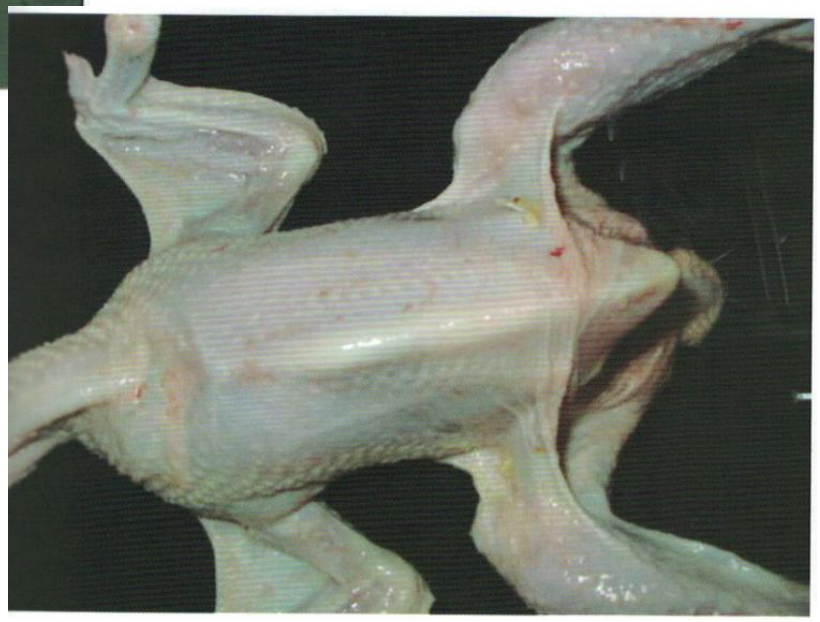


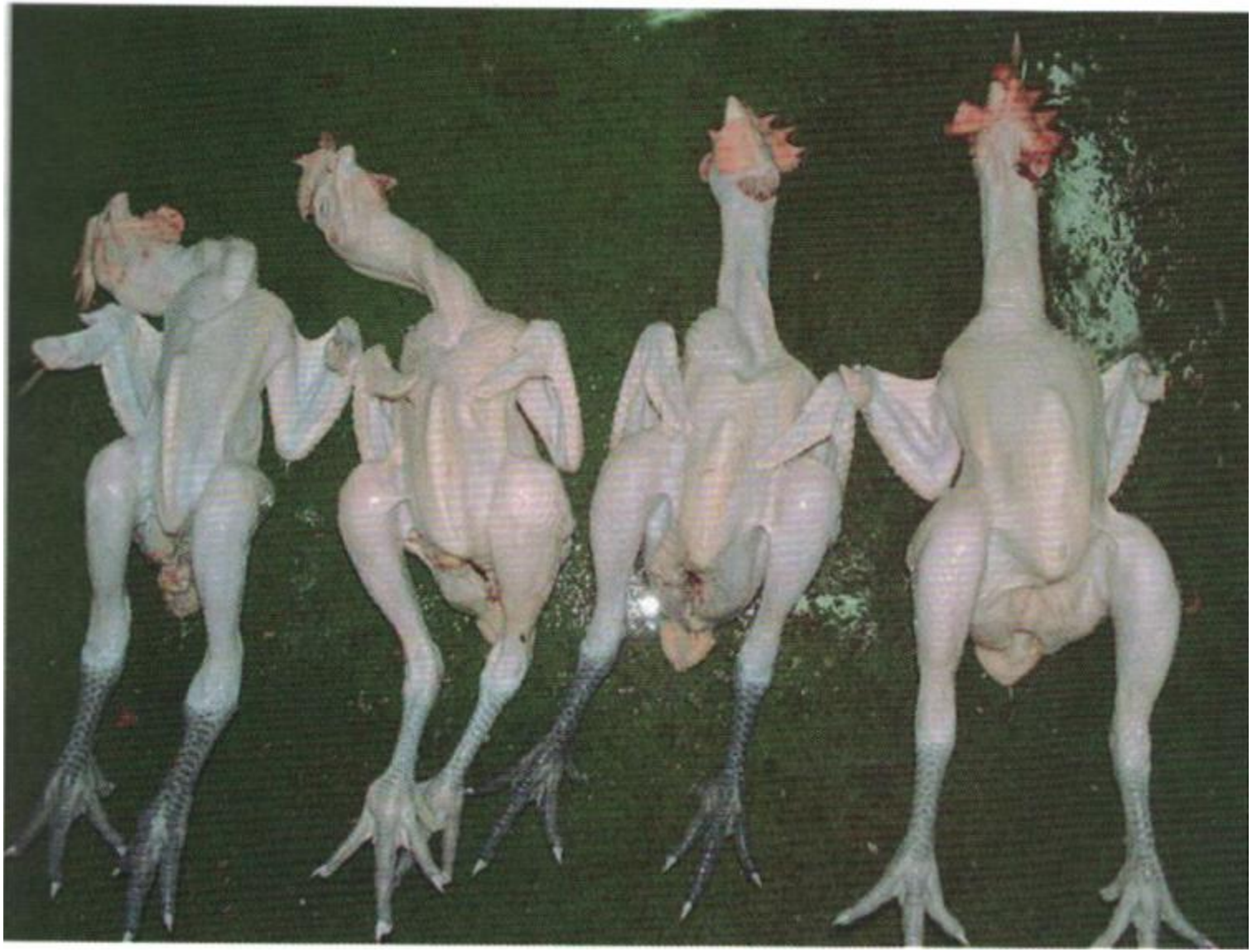
Collect a cloacal swab



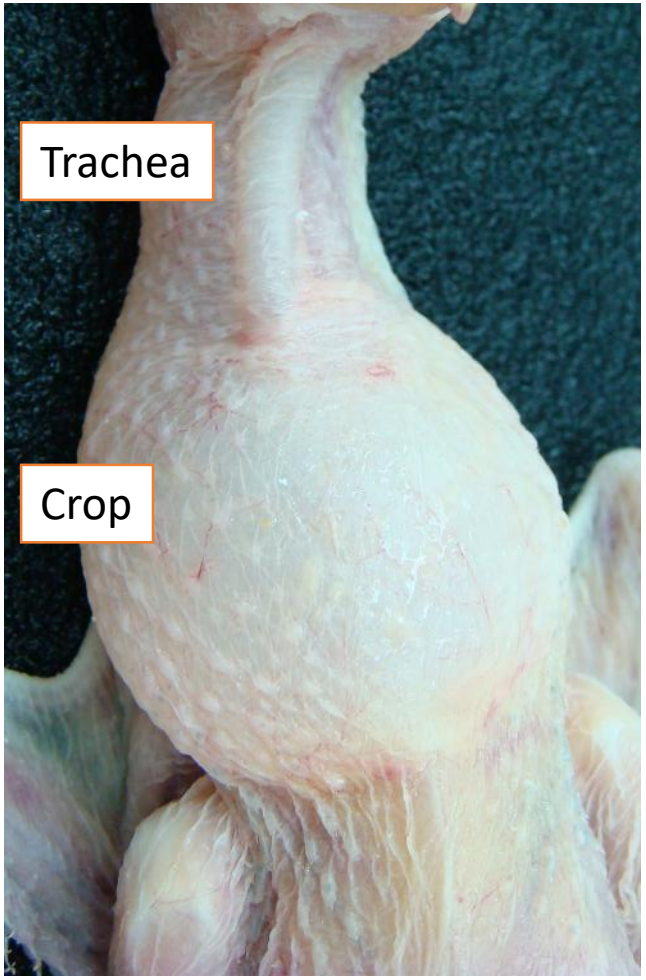
https://www.agric.wa.gov.au/

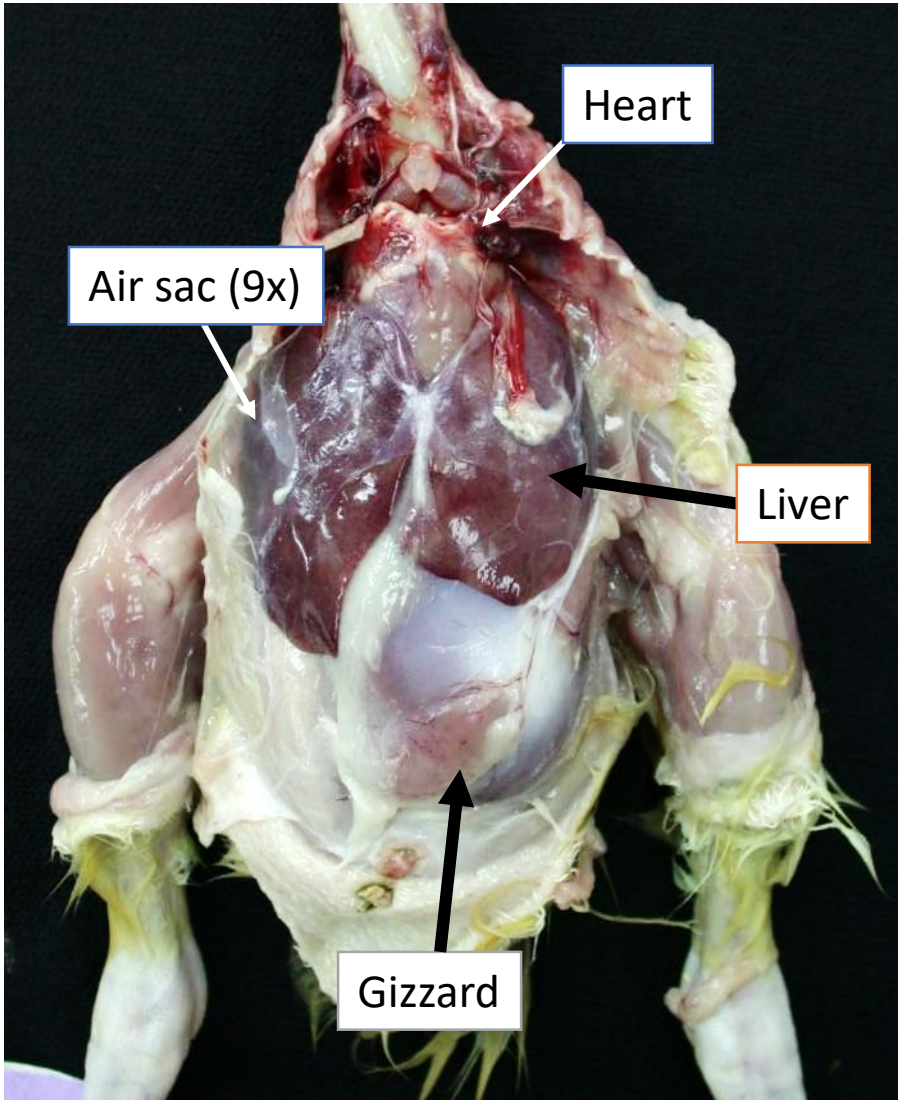




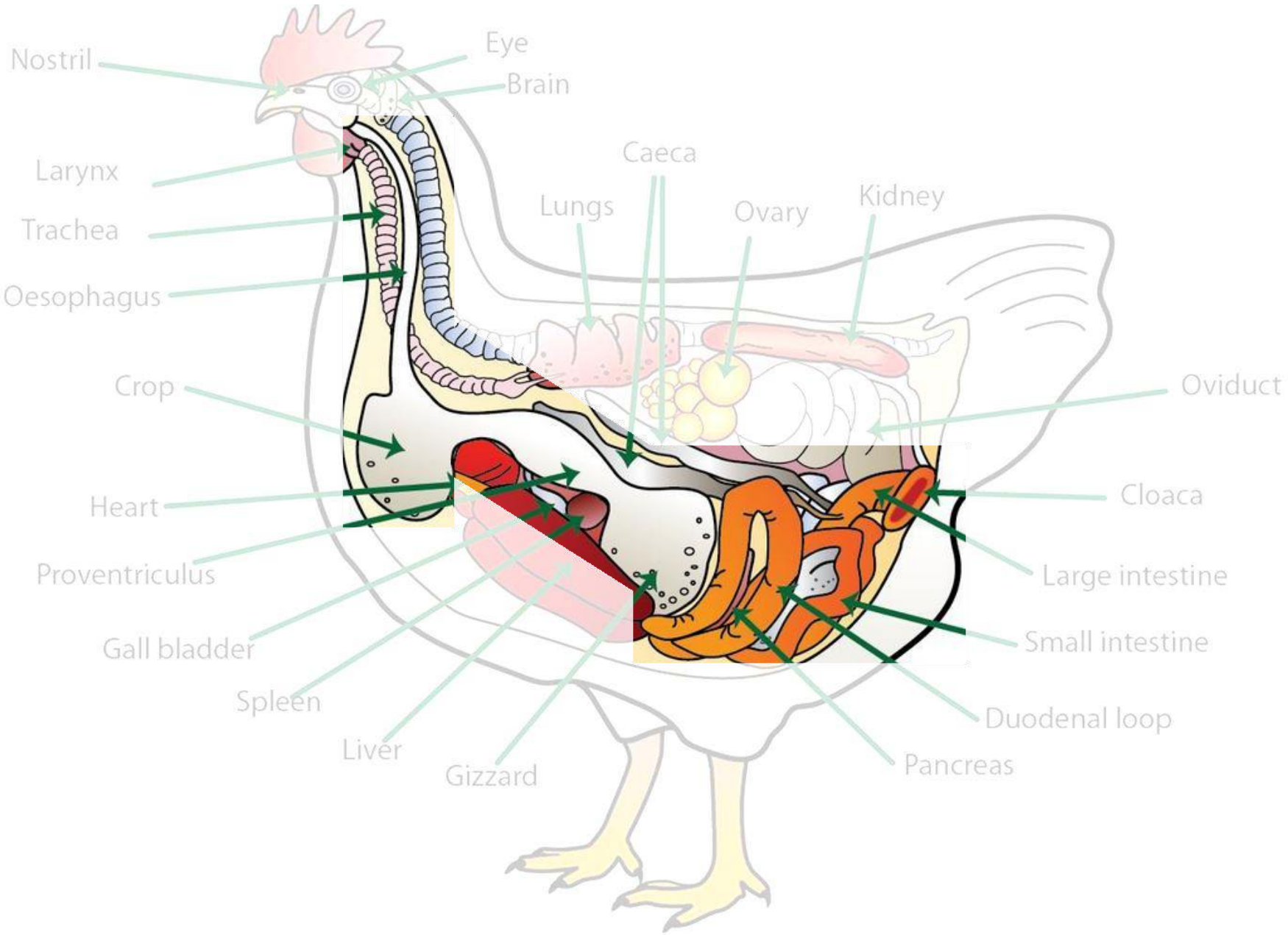


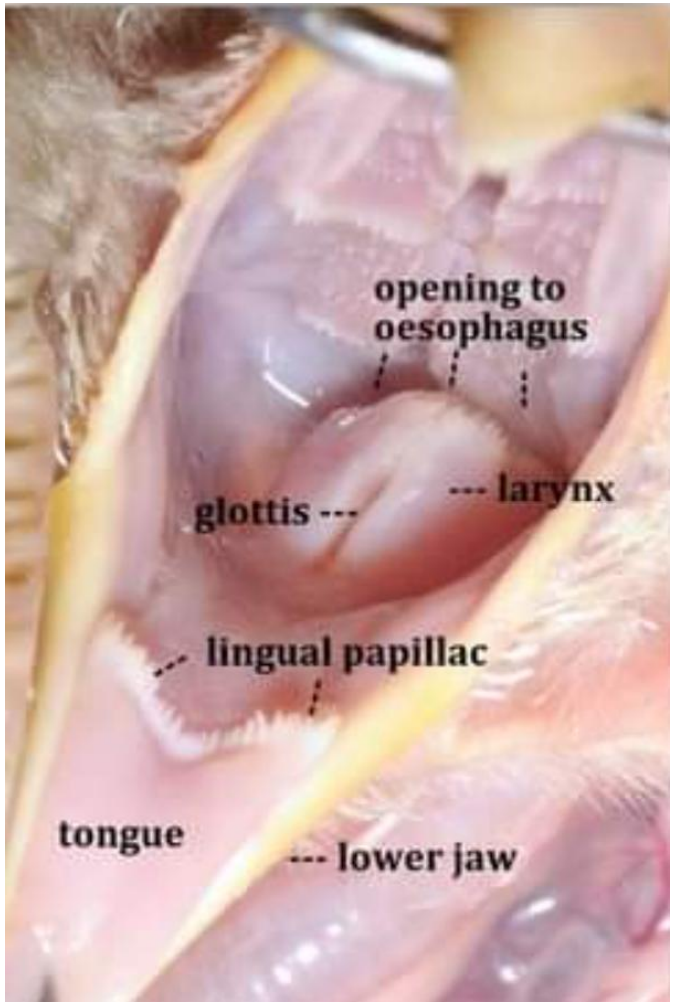
- Anatomy, gross

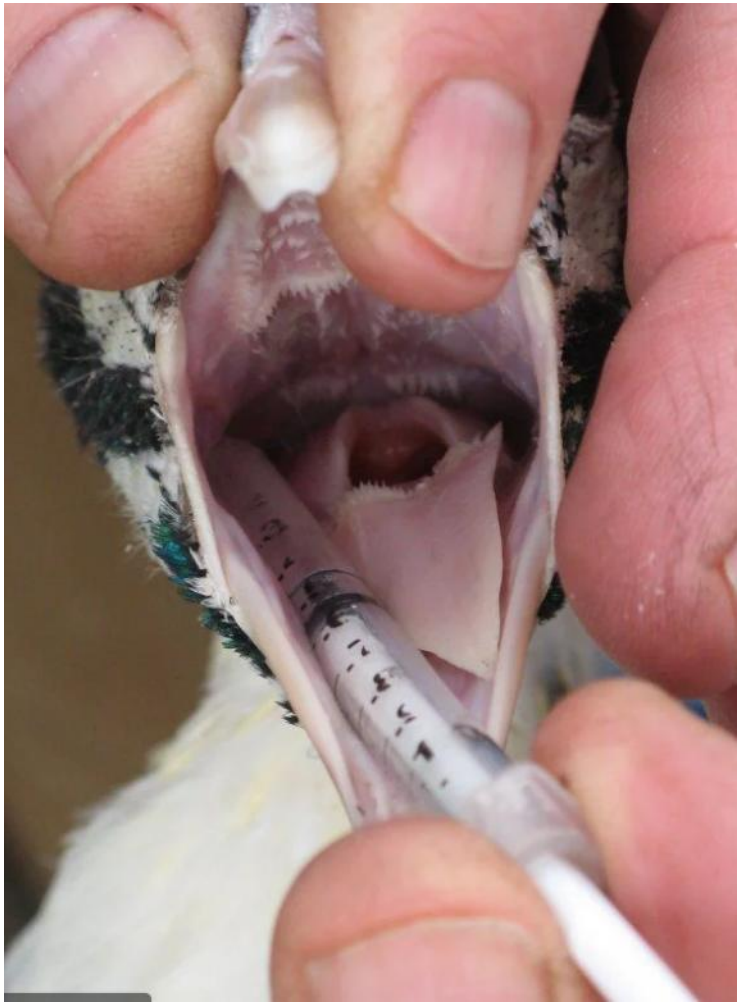


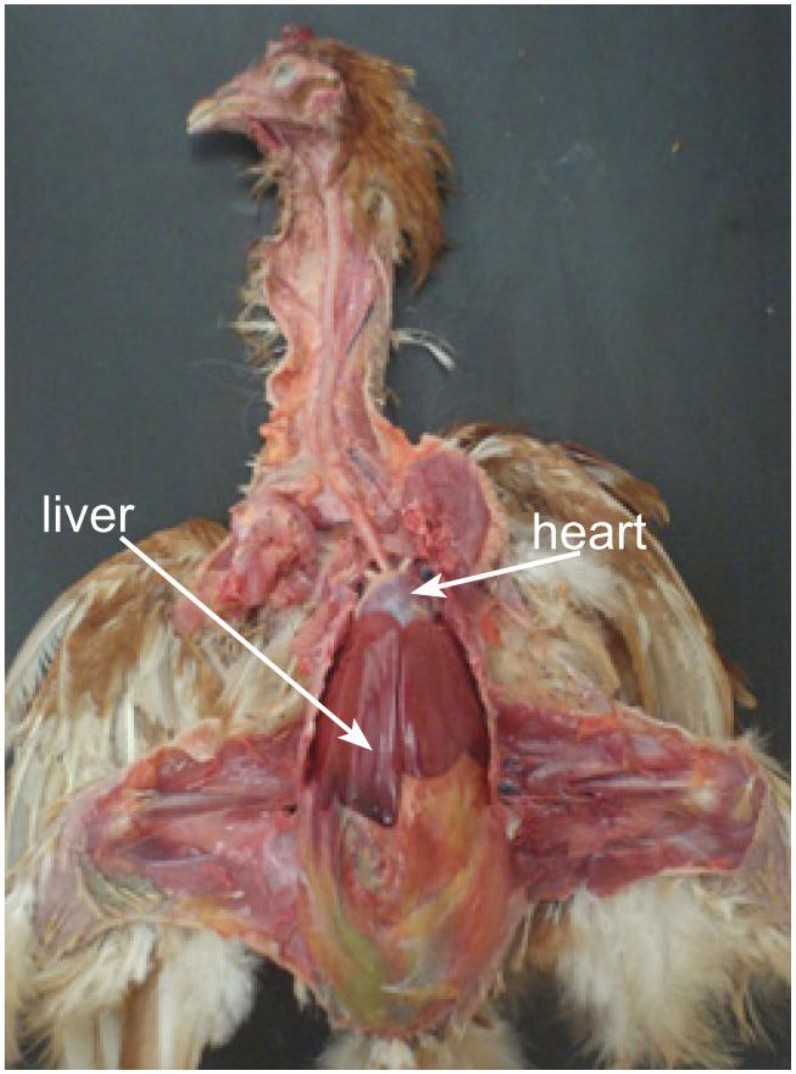


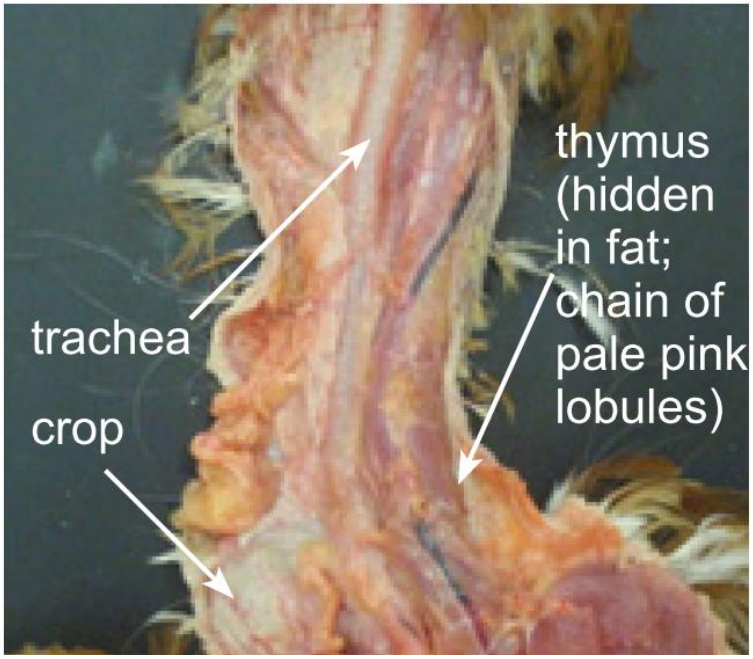
- Digestive system

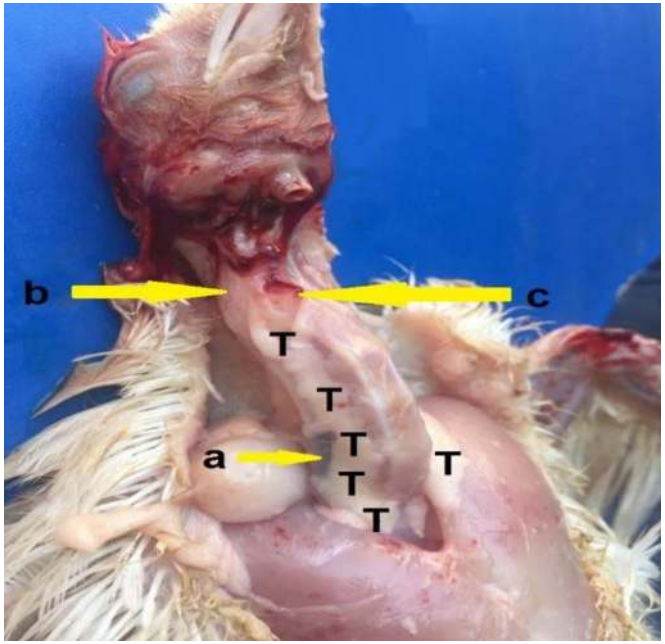



opening to
oesophagus
glottis ---
--- larynx
--- lingual papillac
tongue
--- lower jaw



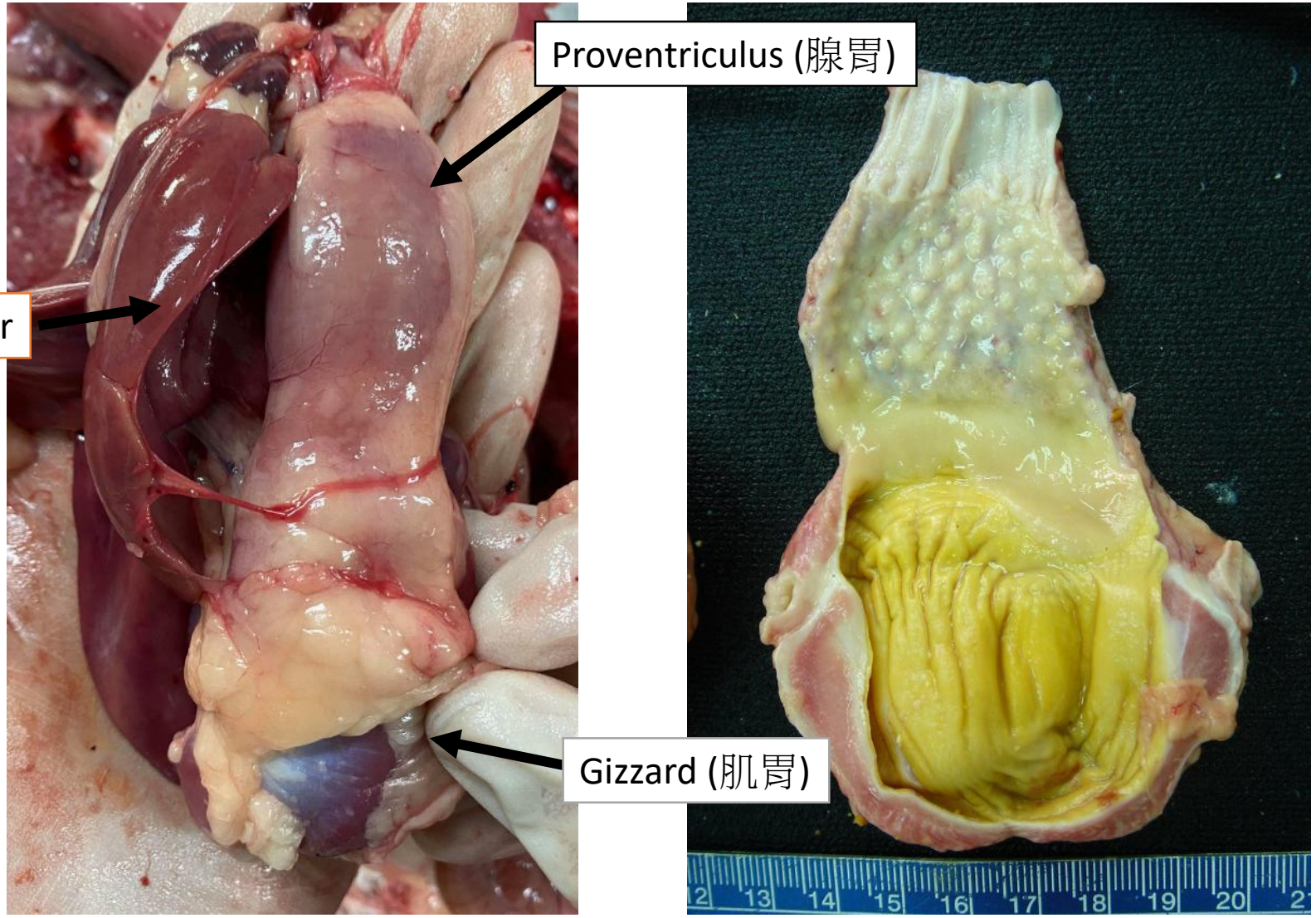

liver
heart


trachea
crop
thymus (hidden in fat; chain of pale pink lobules)



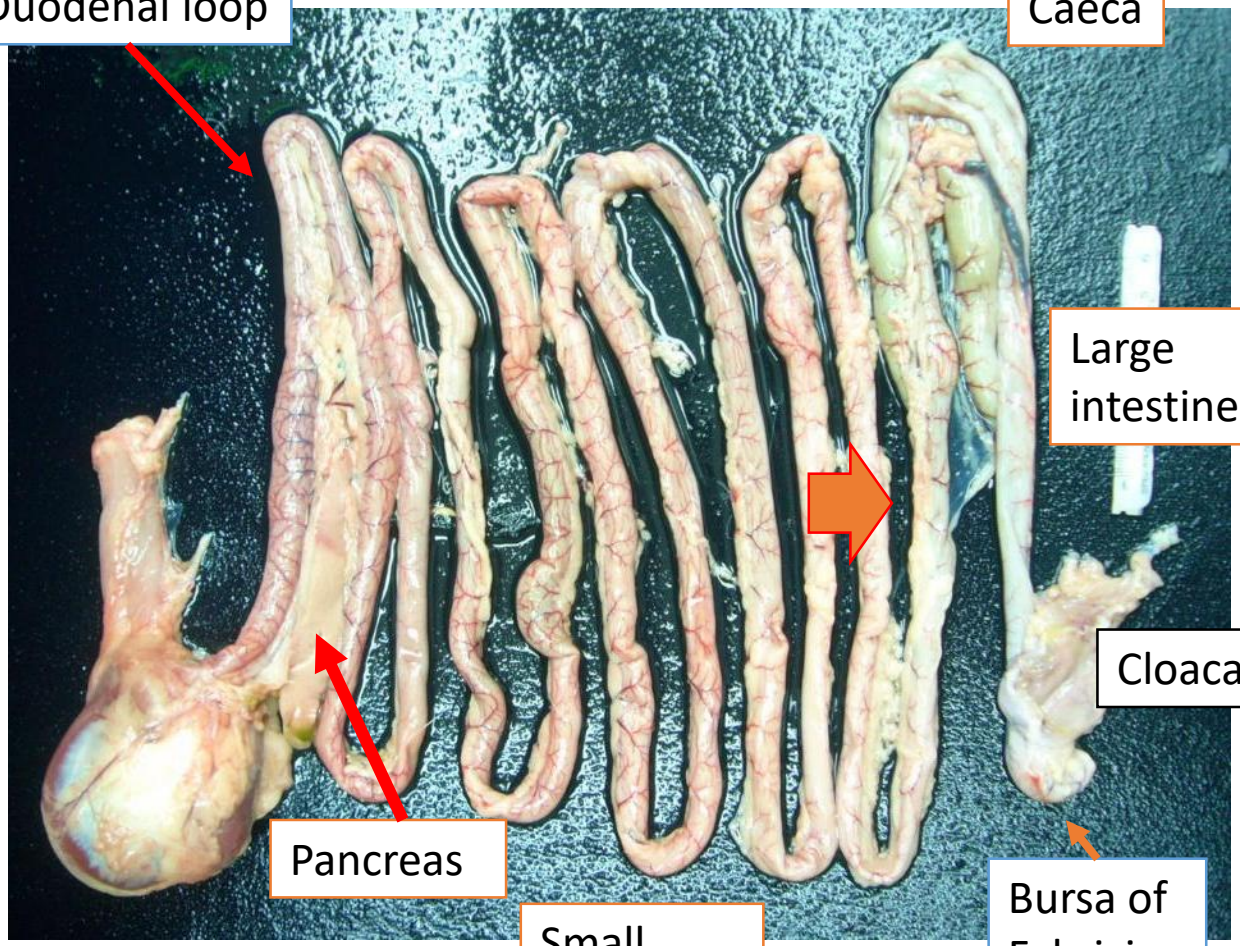
(Alshammary 2017)

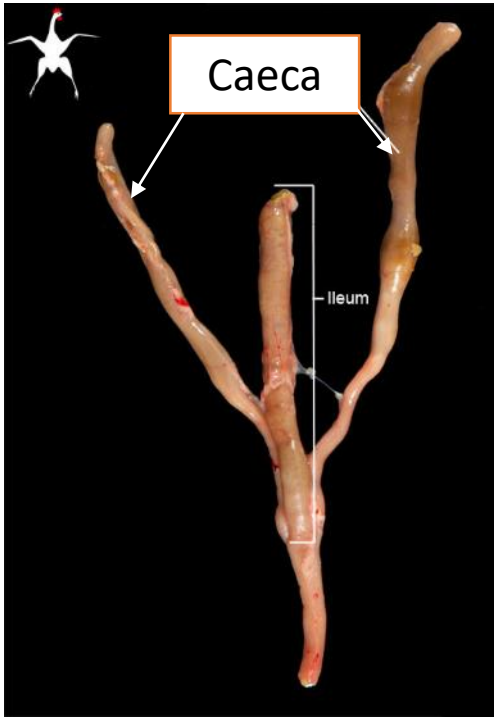
- Anatomy, gross



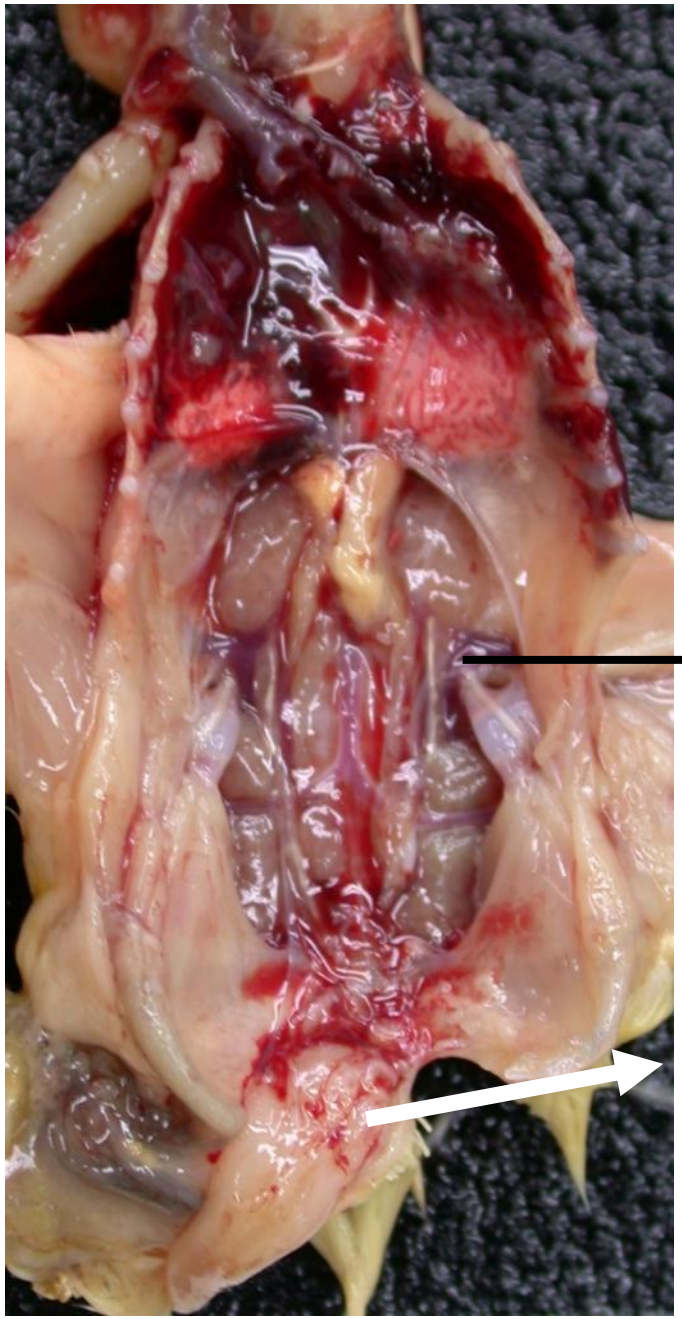
- Digestive system

Ferments left over food and absorbs water

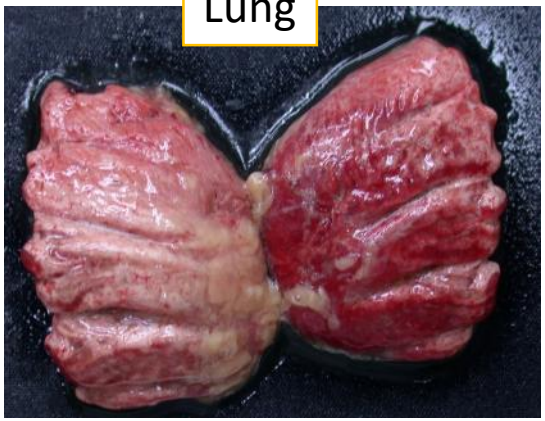




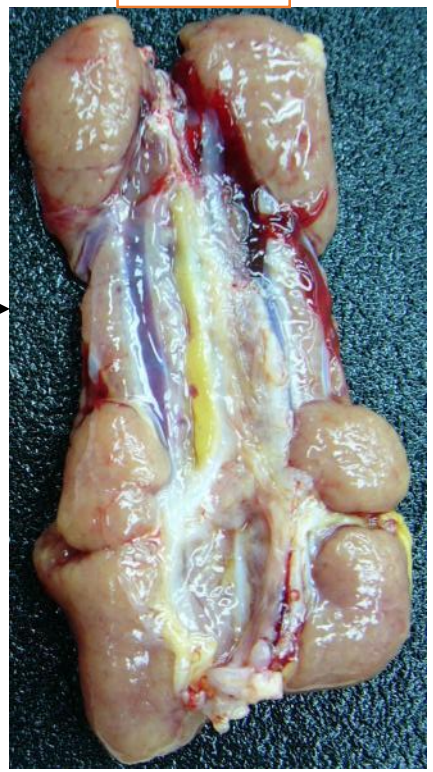
Cornell.edu



Lung



Kidney

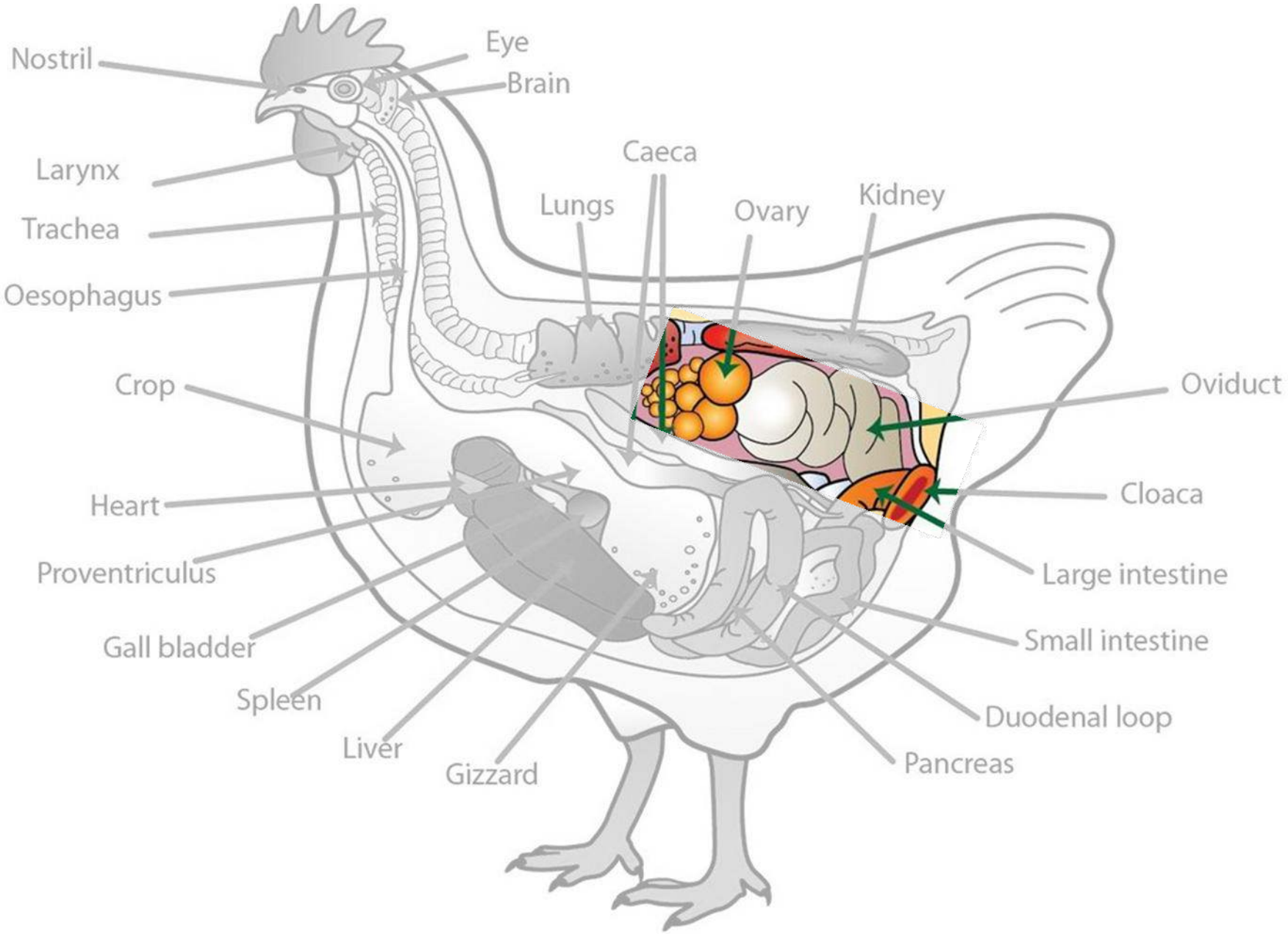




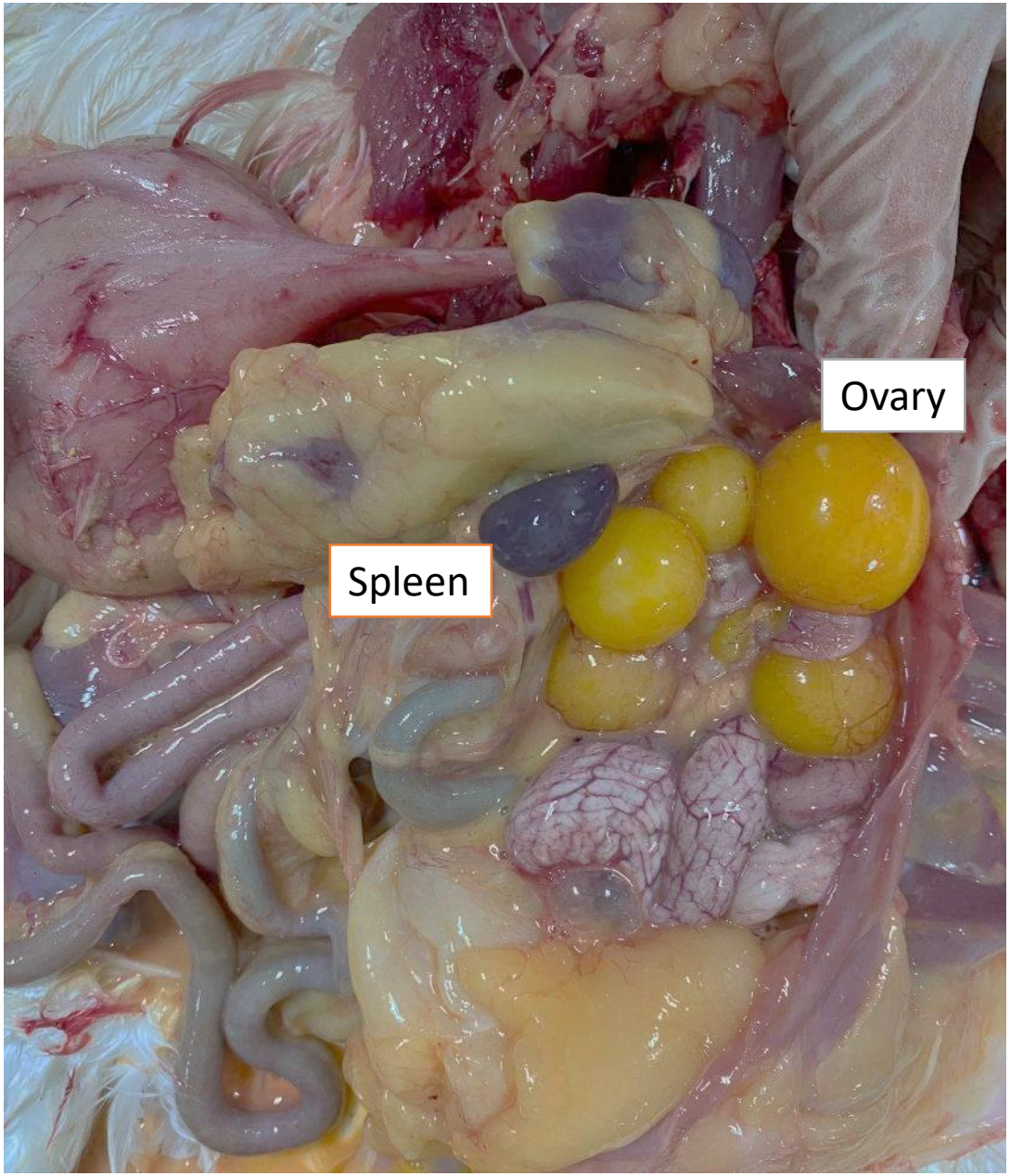
Bursa of Fabricius



- Production system

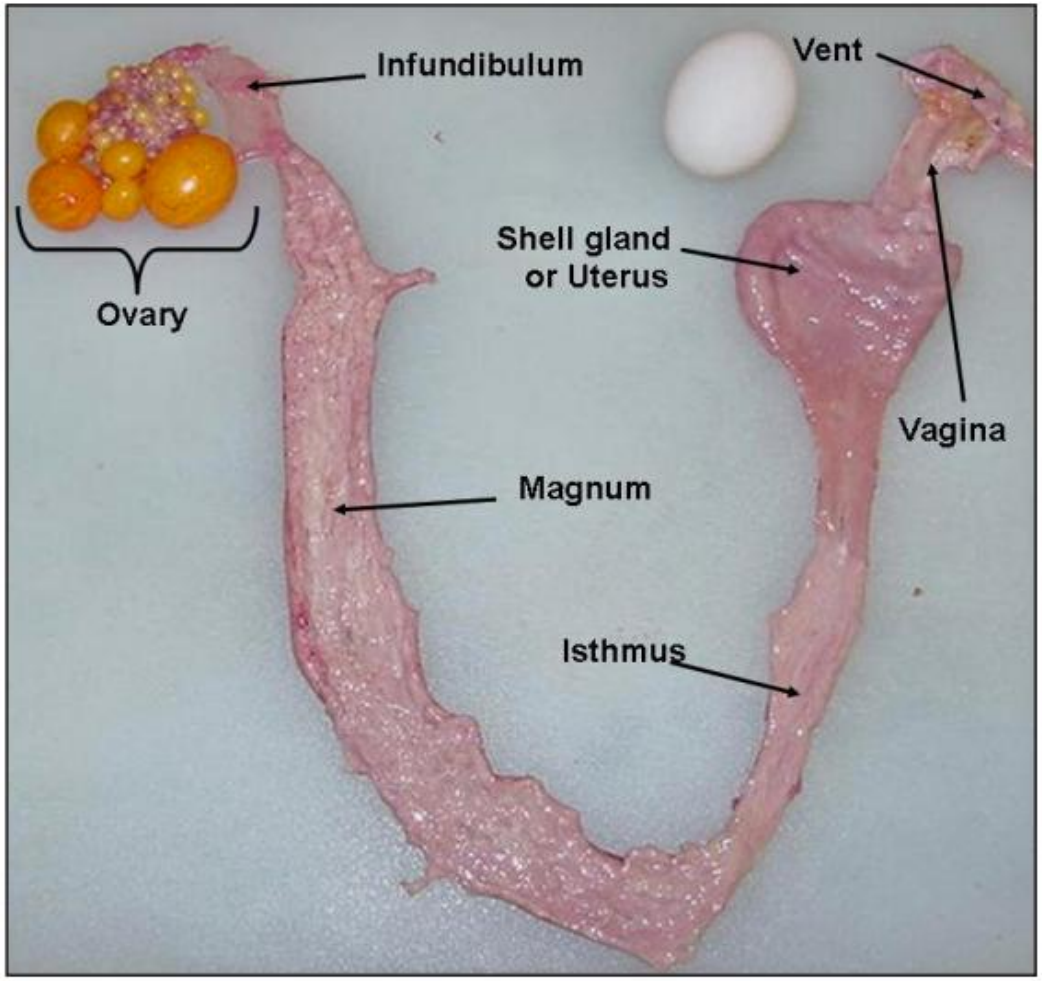


- Anatomy, gross

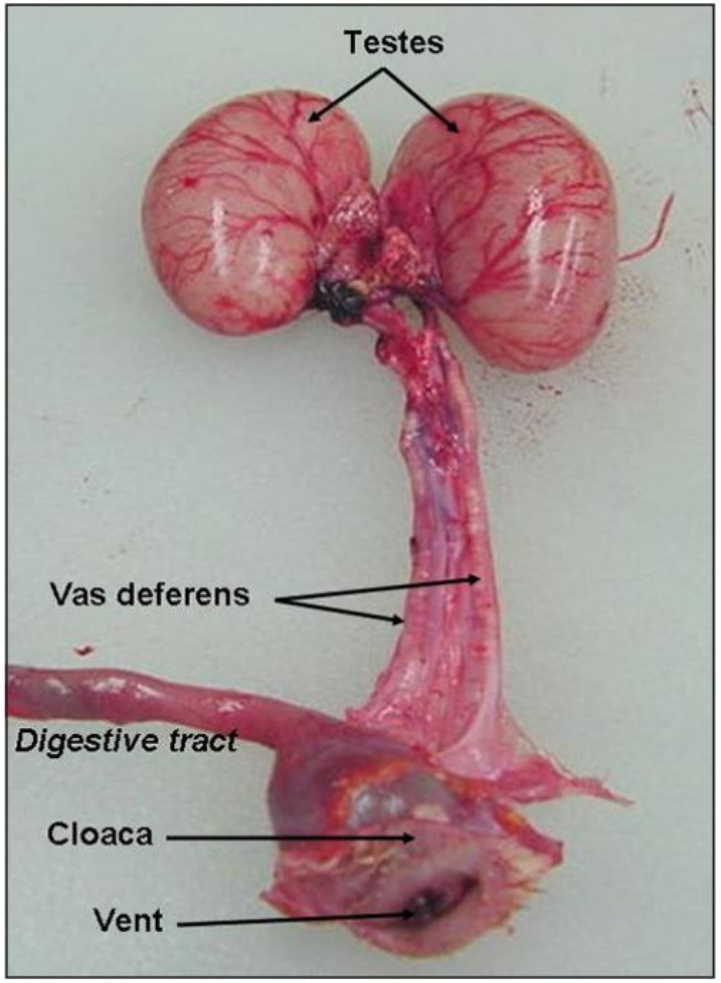


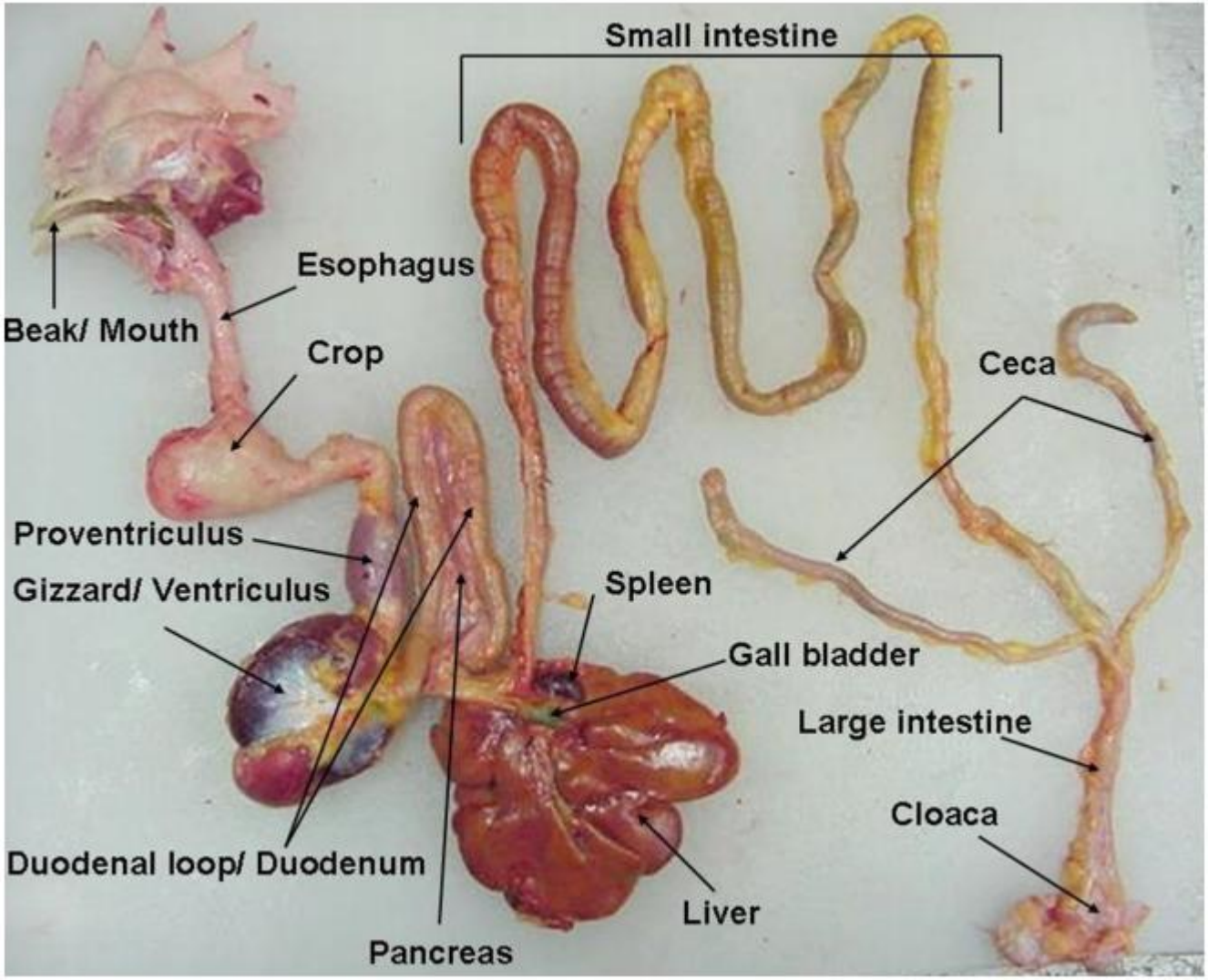
- Reproductive system

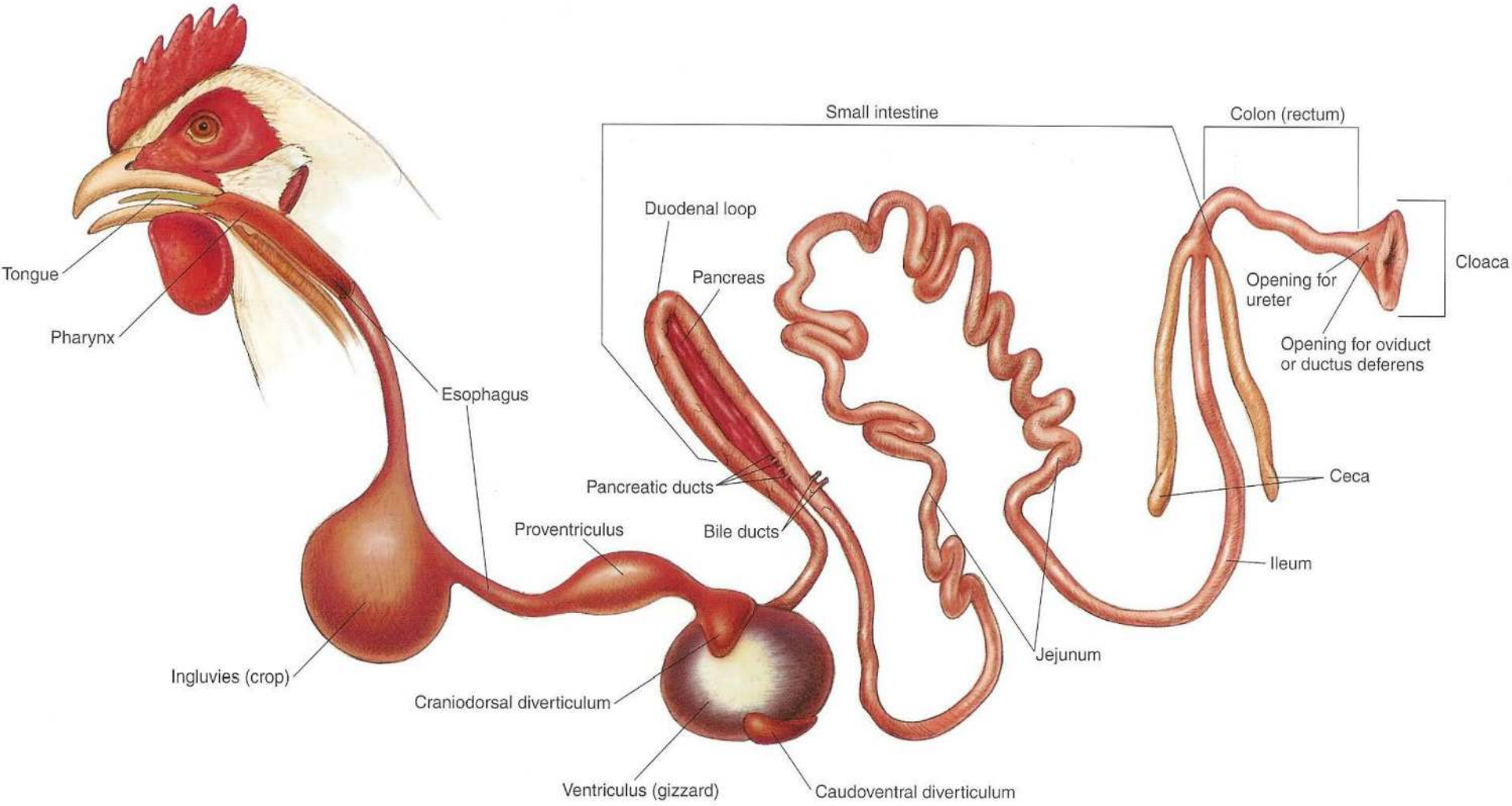
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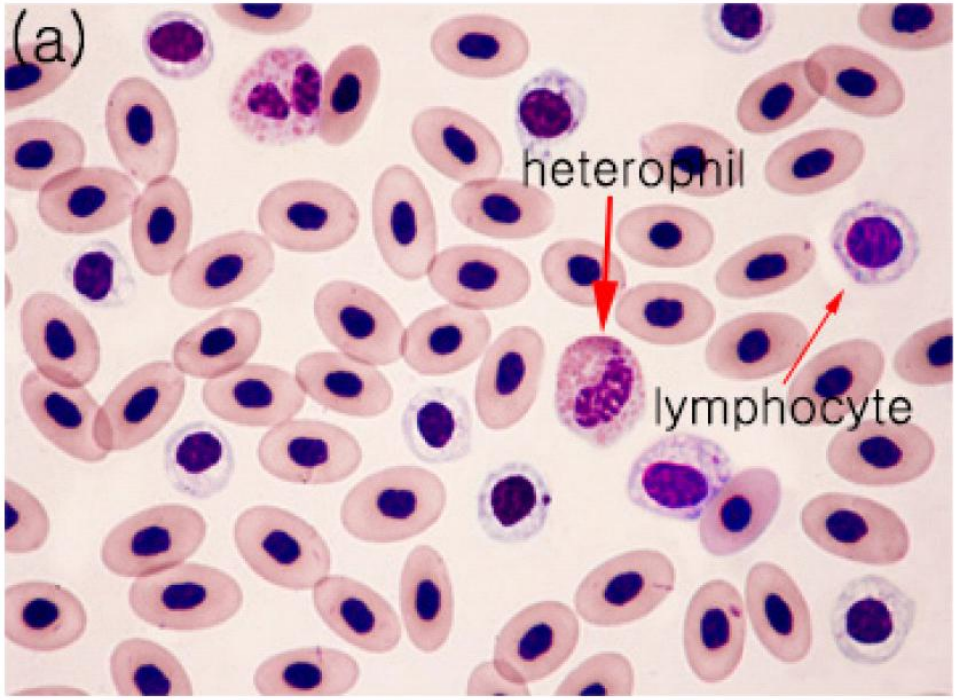


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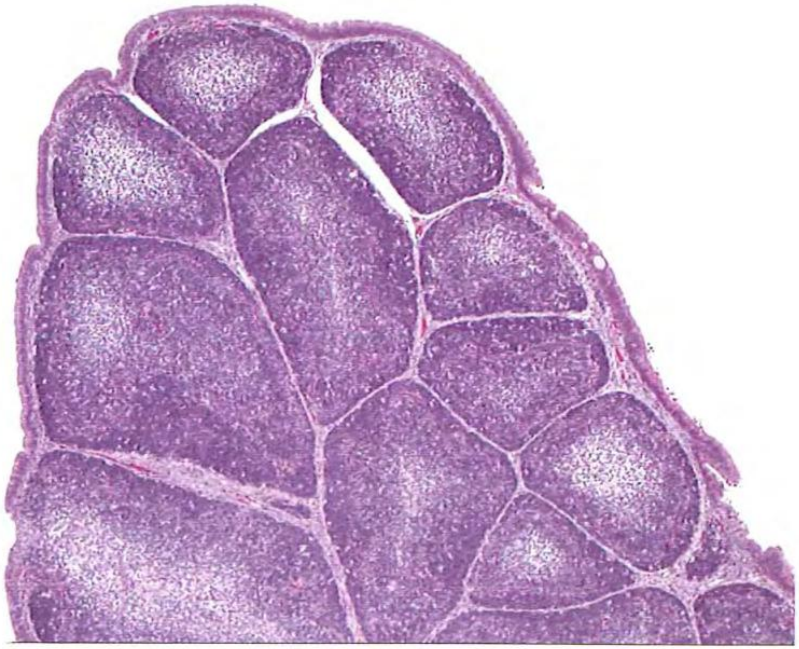


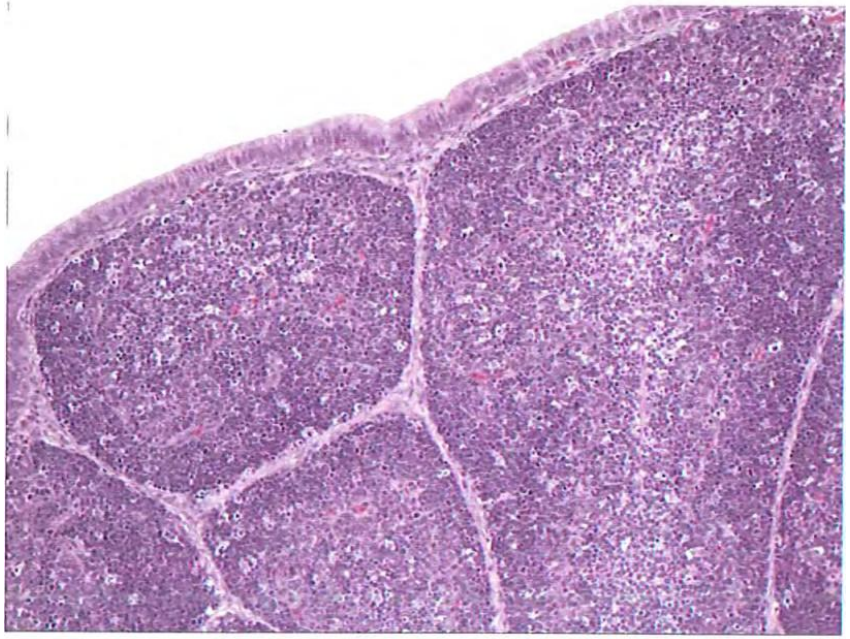

Small intestine
Esophagus
Beak/ Mouth
Crop
Ceca
Proventriculus
Gizzard/ Ventriculus
Spleen
Gall bladder
Large intestine
Cloaca
Duodenal loop/ Duodenum
Liver
Pancreas

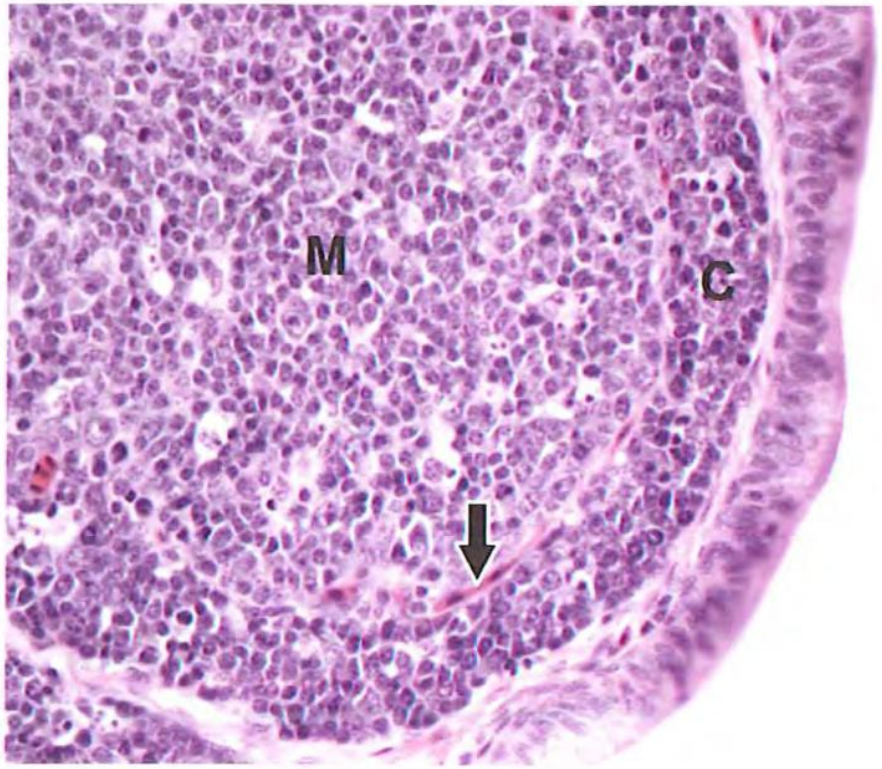

Small intestine
Colon (rectum)
Duodenal loop
Pancreas
Tongue
Pharynx
Esophagus
Opening for ureter
Cloaca
Opening for oviduct or ductus deferens
Pancreatic ducts
Ceca
Proventriculus
Bile ducts
Ileum
Jejunum
Ingluvies (crop)
Craniodorsal diverticulum
Ventriculus (gizzard)
Caudoventral diverticulum



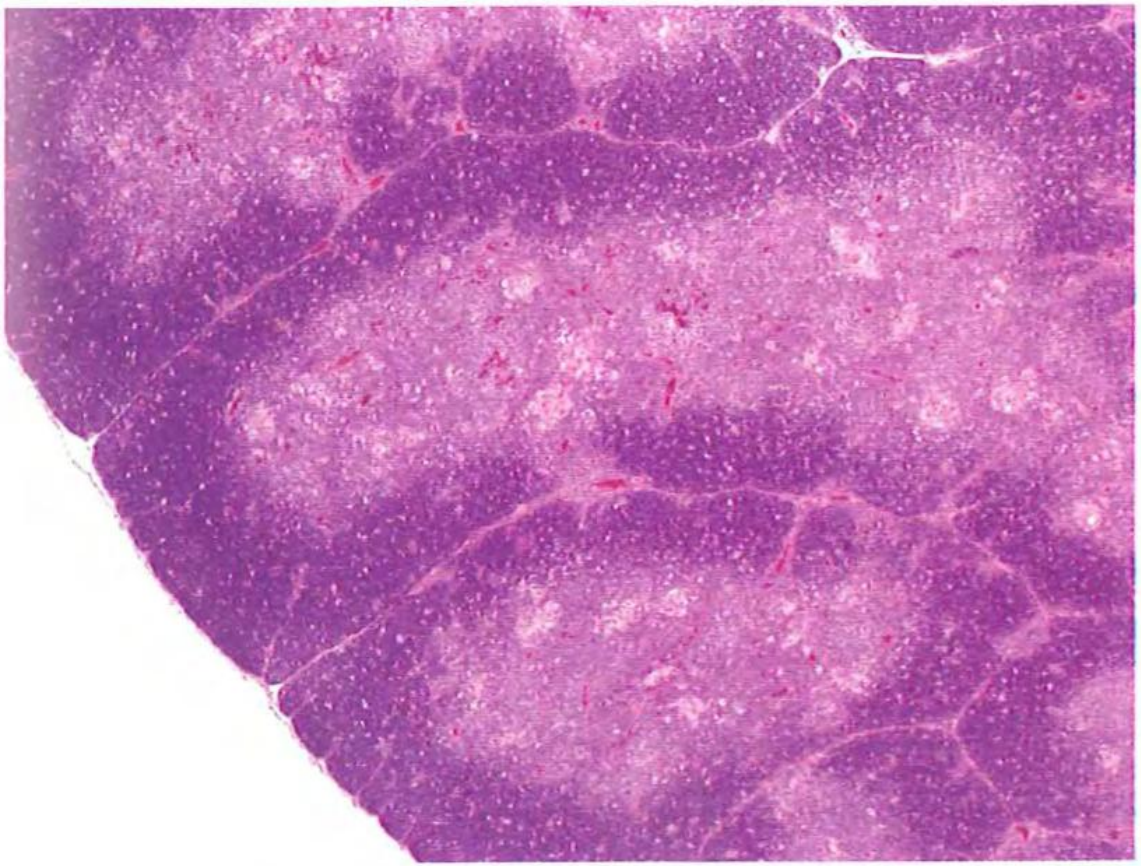
Korean J. Poult. Sci. 2020

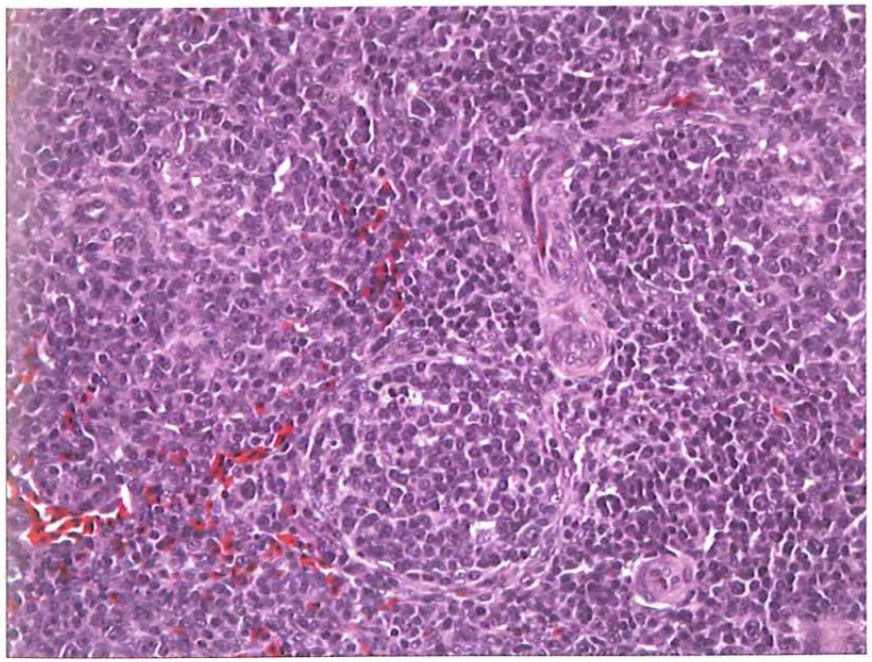


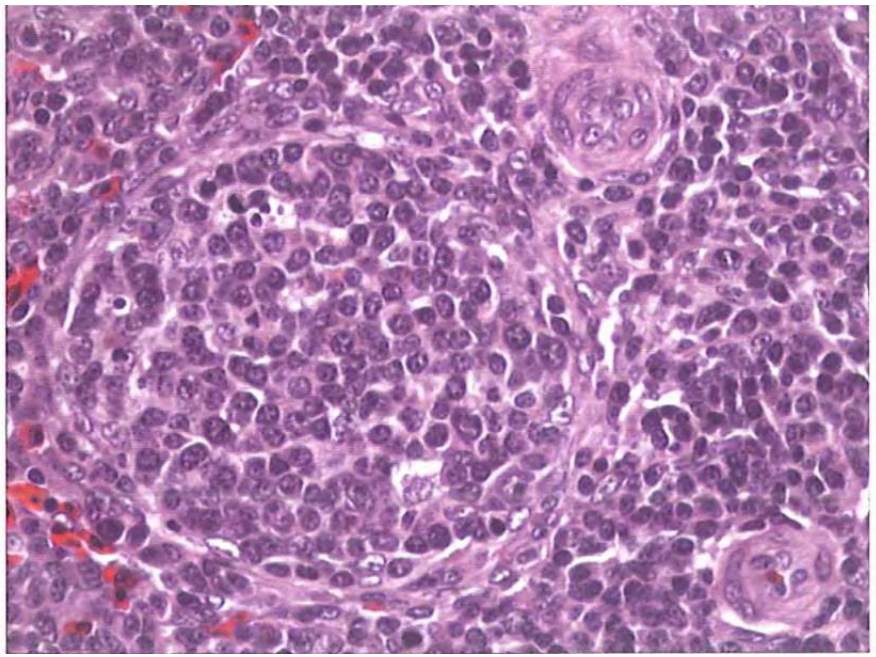


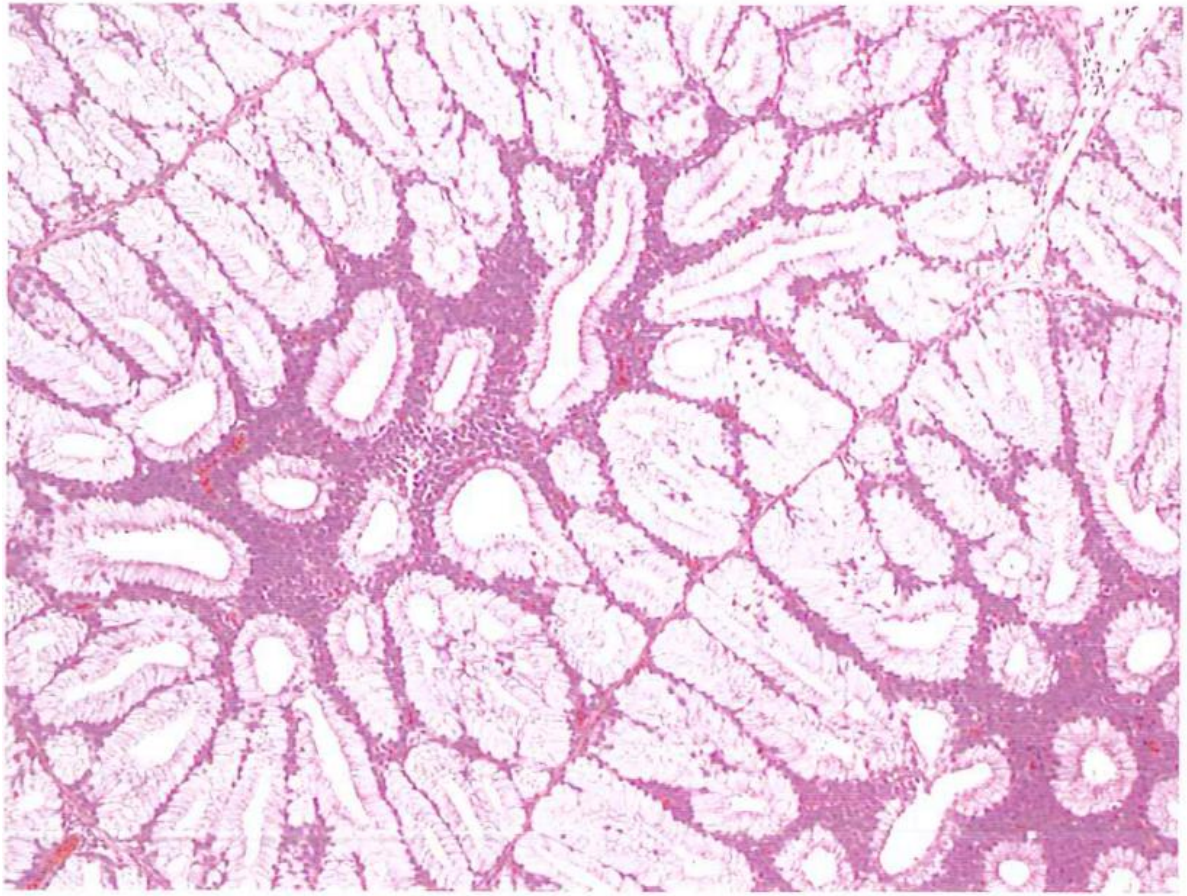


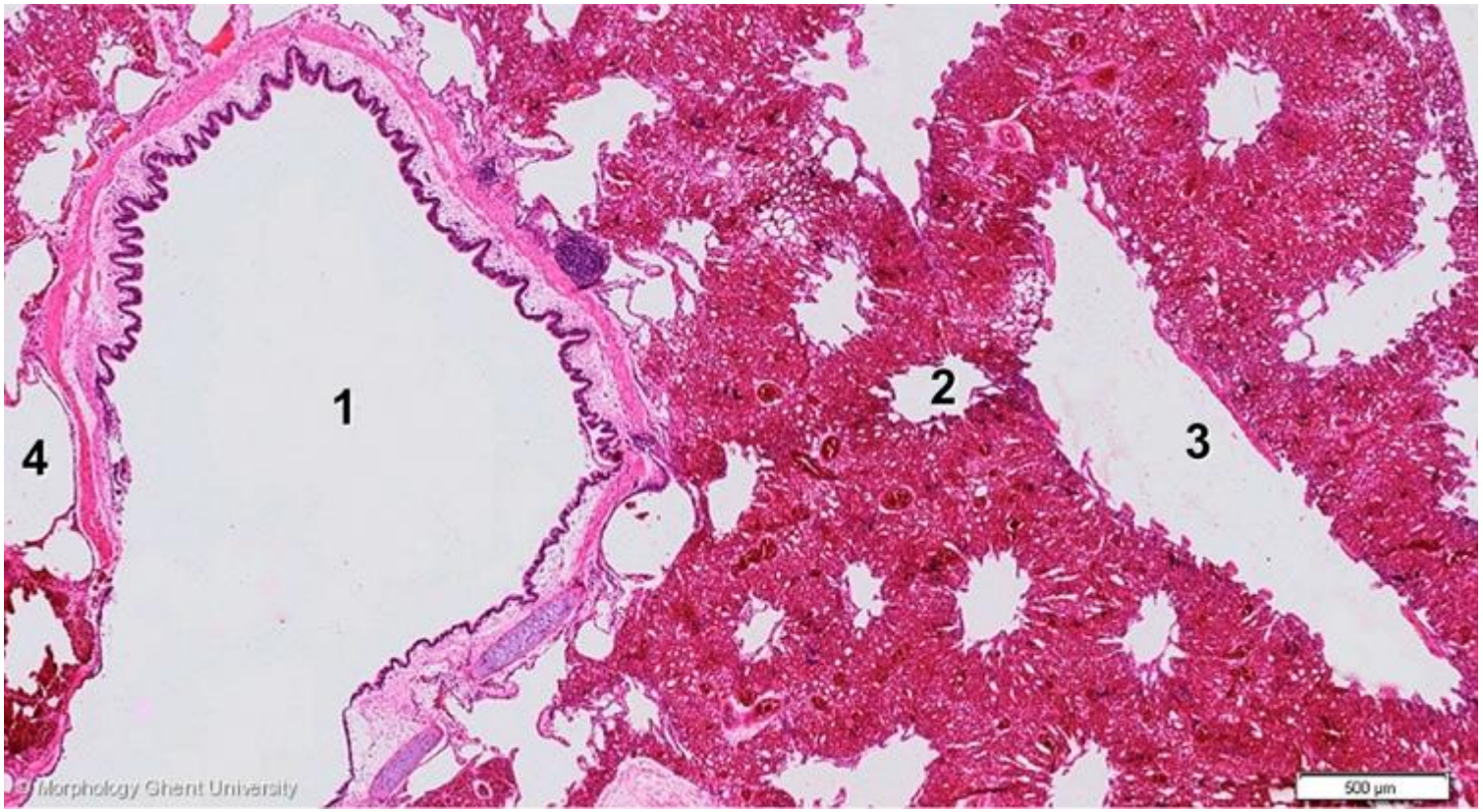
2.4. Bursa of Fabricius. Normal. 3-month-old backyard chicken. Lymphoid follicle shows the cortex (C), medulla (M), and cortico-medullary border (arrow). Note the blood capillaries in the



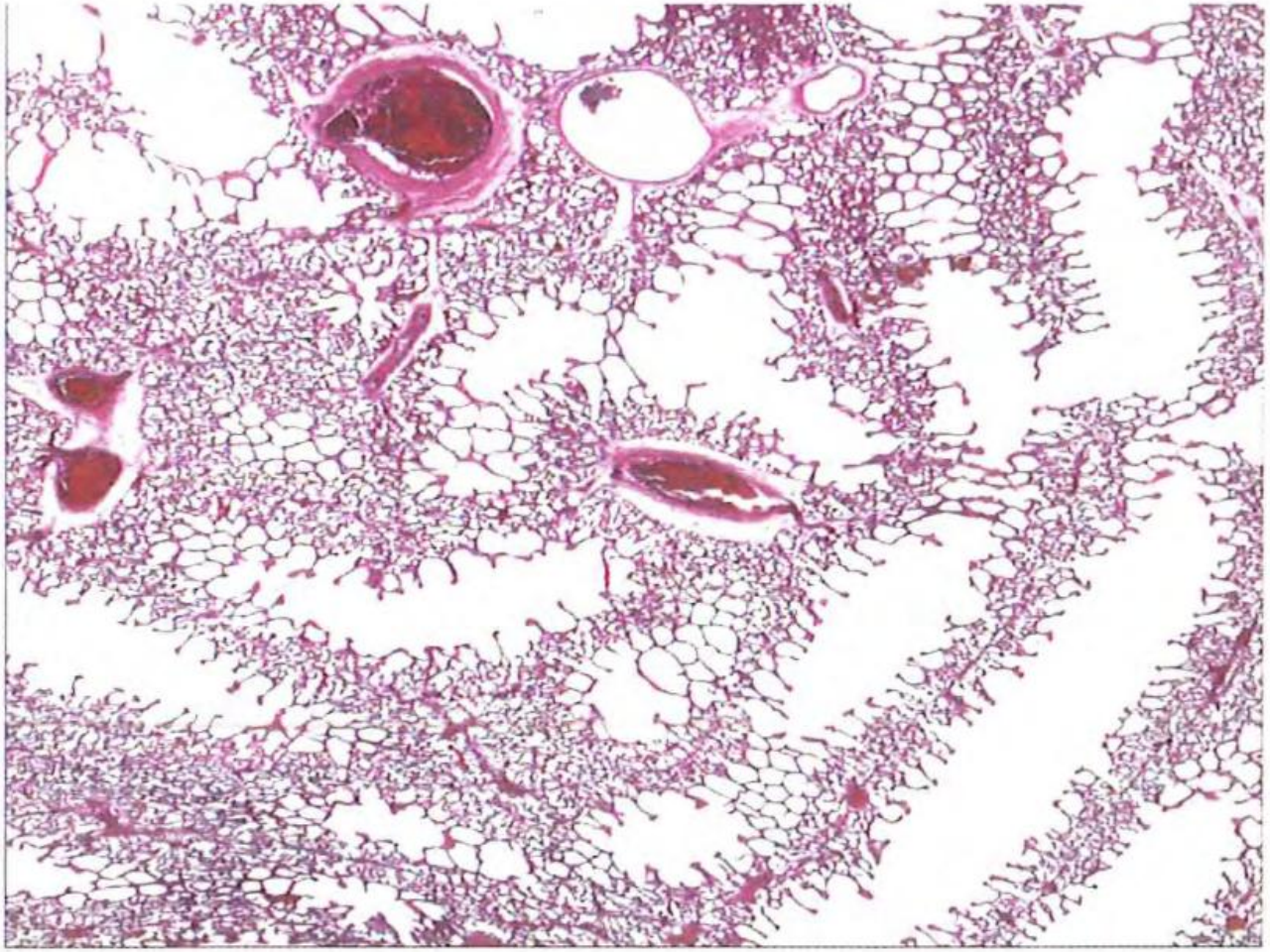








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